Search Incursions: A Relational Theory of Disequilibrating Inventions and Recombinant Advantage

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Abstract

Each novel invention can be viewed as a recombination of existing knowledge elements. By bridging previously less connected elements, an invention changes what constitutes "local" knowledge for firms and reshapes the interfirm matrix of search costs. These changes can enable firms to enter the established inventive territories of other firms; an event we call a search incursion. For each invention (I), we theorize how each firm's position in the technological knowledge network, relative to the invention and other firms, shapes its opportunities to undertake incursions and its exposure to incursions by others. To test our theory, for each focal invention, we reconstructed the technological network of U.S. patents granted during the preceding fifteen years, then introduced the invention and its citations to prior patents, holding the existing knowledge network and firms' prior knowledge positions fixed. Modeling firms as purposeful but error-prone searchers on this network, we formulated each firm's search as a Markov decision process, and employed policy iteration to estimate how the invention changed the minimum expected cost of searching from each firm's knowledge position toward that of every other eligible firm. Calculating the invention-induced changes in the interfirm search-cost matrix allowed us to estimate the relative recombinant advantage conferred by the invention upon each firm. Separately, we used patent text and synthetic-patent constructions to identify subsequent search incursions, finding support for our predictions concerning which firms undertook incursions, which firms became susceptible to them, and which particular firm-to-firm incursions materialized.

Introduction

The firm Toyobo traces its origins to a cotton-spinning enterprise established in Japan in 1882. By the 1980s, the company had become a manufacturer of synthetic fibers, polymer films, plastics, and other industrial materials (Toyobo, n.d.). Another Japanese firm, Hitachi, began in 1910 as a repair shop for mining machinery, and had subsequently developed into a major manufacturer of electrical and electronic equipment (Hitachi, n.d.). By the 1980s, Hitachi's substantial inventive activities included the development of image-processing technologies which enabled computers to recognize the shapes and positions of objects in images (US 4,491,962).

In 1987, Konica, a firm specializing in photography and imaging, received a patent for an invention introducing a new method for converting images into black-and-white representations while preserving information important for recognizing their contents (US 4,710,822). Image processing of this

nature also appears to have had practical relevance to Toyobo's established manufacturing activities. For instance, optical inspection technologies could, among other potentially useful applications related to Toyobo's main areas of inventive focus, identify defects in woven fabrics. By 1990, Toyobo had developed and patented a novel method for processing images while preserving the information needed to recognize shapes (US 5,296,940). Toyobo in this case appears to have built upon Konica's invention, combining elements of this new technological knowledge alongside its own accumulated expertise, to develop a novel invention in an inventive domain where Hitachi had previously enjoyed inventive pre-eminence.

We conceptualize the act of one firm producing an invention in a domain characterized by another firm's substantial inventive activity, as a *search incursion*[1]. Formally, we define search incursions as cases when a firm produces a novel invention outside the regions of inventive space in which it has historically concentrated its inventive activity, and within a region in the broader inventive space more firmly associated with the prior inventions of other firms.

This motivates the central question of our study: **as a consequence of the arrival of each particular invention I, can we explain which firms A, B, and C become better positioned to undertake search incursions? Can we further understand which firms become more vulnerable to incursions by others? And finally, can we pin-point to which particular firms' established inventive regions have become susceptible to become the targets of each well-placed firm?**

Understanding the conditions that enable search incursions, and also the attributes of the firms conducting (or being subject to) incursions has rich theoretical and practical implications for at least the following two significant reasons:

First, since patents or other intellectual property arising from an inventive incursion may introduce competing claims that constrain the ability of firms already active in the technological area to appropriate value from their prior inventive efforts (Ziedonis, 2004), the occurrence of a search incursion can constitute a consequential episode in the competitive relationship between the firms involved. However, even where intellectual property rights may not be of substantial concern, the development of inventions by another firm within an established inventive region can constrain the strategic autonomy of the affected firms. For example, decisions concerning commercialization, as well as the direction and timing of subsequent inventive efforts, may now have to account for the technological and competitive actions of the incursing firm (Stuart and Podolny, 1996; Katila and Chen, 2008). For Hitachi therefore, Toyobo's apparent search incursion could have consequences extending beyond the immediate technical contribution of Toyobo's invention, potentially having far reaching effects on Hitachi's subsequent inventive and commercialization choices.

Second, viewed from a broader ecological lens, a search incursion can be understood as one prominent competitive *micro-interaction* through which broader technological transformations emerge.

[1] Please note that this spatial characterization would appear to be of low resolution. In practical terms, this would mean that a firm has developed an invention that embodies functionalities or attributes that would characterize another firm's invention. We use a spatial characterization for both an ease of explanation, and also because a high resolution approach would distract from the main thrust of the theory, which we develop in the later sections of this paper

From ideas such as technological convergence (Rosenberg, 1963; Uzunca, 2018) to notions of discontinuities that destroy established technological competencies (Tushman and Anderson, 1986; Rothaermel and Hill, 2005), the strategy literature has developed substantial accounts of changes in the competitive and inventive landscape across firms. Such transformations can be viewed as aggregations of pairwise changes in the inventive activities of firms, of which a search incursion could be a principal variant.

Understanding these micro-interactions can therefore provide a highly resolvent basis for recognizing broader technological transformations while they are still in the course of formation. Consider the following analogy: whereas documenting the aggregate consequences of a pandemic reveals its severity and populational consequences, understanding how infection passes between individuals provides the atomic information necessary for each person to assess their own exposure and take appropriate measures to protect themselves. Analogously in our case, understanding how each particular invention changes the prospects for each particular firm to enter another firm's established inventive regions can enable the concerned firms to recognize emerging opportunities (or threats), before their broader competitive consequences have fully materialized. A micro-interactional account of firm interactions could therefore be an important driver of firm-specific strategic decisions about which inventive opportunities to pursue, which established positions to defend, and how to respond as technological change unfolds.

Several existing studies in the strategy literature provide important foundations for understanding our lines of investigation. Studies on technological change have explained how different technological developments reshape competitive advantage, including how competence-enhancing and competence-destroying discontinuities, architectural innovations, and changes in complementary assets affect different classes of firms (Tushman and Anderson, 1986; Henderson and Clark, 1990; Tripsas, 1997; Rothaermel and Hill, 2005). On the other hand, important research on innovation search, meanwhile, has examined how firms' choices of knowledge components, patterns of exploration, and search practices shape their inventive and product outcomes (Ahuja and Lampert, 2001; Rosenkopf and Nerkar, 2001; Katila and Ahuja, 2002). Related studies have also characterized firms' positions in technological space and examined how those positions evolve through inventive search (Jaffe, 1986; Stuart and Podolny, 1996; Podolny, Stuart, and Hannan, 1996). Taken together, these research streams have provided substantial explanations of the consequences of technological change, the determinants and outcomes of inventive search, and the technological positions firms occupy. However, these studies leave unexplored the specific question of (a) how each particular invention changes the prospects for each particular firm to undertake inventive activity in a region previously associated with another firm, and (b) how these inventions redistribute opportunities for incursive search and exposure to incursions across particular firms.

To answer these questions, we develop and test a structural theory of how individual inventions redistribute inventive opportunities and vulnerabilities across firms, advancing toward a firm-by-firm, invention-by-invention explanation of technological competition.

Our explanation draws on three complementary theoretical foundations. First, research on recombinant innovation conceptualizes invention as a process of searching over combinations of existing knowledge components, with the feasibility and costs of recombination depending on the relation-

ships among those components (Fleming, 2001; Fleming and Sorenson, 2001, 2004). Second, research on technology and technological change establishes how new inventions can alter the technological feasibility and economic attractiveness of subsequent inventive activity (Dosi, 1982; Bresnahan and Trajtenberg, 1995). Third, research on organizational learning and local search provides a basis for understanding how firms' accumulated knowledge and technological positions condition the effort required to pursue different inventive possibilities (Cyert and March, 1963; Cohen and Levinthal, 1990; Stuart and Podolny, 1996).

Building on these foundations, our theory proceeds as follows. Consider four firms, A, B, C, and D, whose prior inventions are concentrated in different regions of inventive space. An invention (I), developed by a fifth firm, E, may disturb the established distribution of inventive advantages among these firms through at least two mechanisms. First, E's invention may enable A to combine knowledge components that could not previously be combined, allowing A to develop a novel invention within a region of inventive space historically associated with B's prior inventive activity, thereby undertaking a search incursion. Second, the same invention may allow D to bypass a technological component on which C's established inventive advantage depends, making a search incursion into C's established inventive region economically attractive even where such inventive activity was previously technologically feasible. In the first case, the invention opens a previously infeasible recombinant possibility; in the second, it weakens a technological barrier that had previously discouraged inventive entry.

These mechanisms together establish how an invention can make search incursions technologically feasible or economically attractive. However, to explain how the resulting opportunities are distributed across firms, we consider the same invention from another analytical perspective: as disequilibrating forces that alter the network of technological knowledge relationships. Each invention can itself be understood as a recombination of antecedent knowledge components. Therefore, in establishing new connections among knowledge components that were previously separated, E's invention can be seen as changing the structure of the technological knowledge network and, consequently, what constitutes ”local” search from the knowledge positions occupied by each different firm. These new connections alter the effort required for each firm to develop inventions in regions historically associated with other firms' inventive activity. Because firms occupy different positions relative to the connections introduced by the invention, the resulting changes in search costs are neither uniform across prospective destinations nor necessarily equivalent in opposite directions. For example, A may experience a substantial reduction in the effort required to invent in B's established inventive region, without experiencing comparable reductions in the costs of inventing in C's or D's regions. Nor must B experience an equivalent reduction in the cost of inventing in A's established inventive region.

From the perspective of any particular firm, these changes in interfirm search costs have two distinct consequences. First, reductions in the costs of searching toward other firms' established inventive regions improve the focal firm's *recombinant opportunity*, or its ability to pursue inventive activity in those regions. Second, reductions in the costs other firms face in searching toward the focal firm's established inventive region increase its *recombinant exposure*, or its susceptibility to incursions originating from those firms. Crucially, the same invention may improve a firm's recombinant opportunity and increase its recombinant exposure by different proportions. Because pursuing new inventive opportunities and preserving established inventive positions both demand inventive

resources, the relative magnitude of these changes can be the determining factor in governing a firm's propensity to conduct a search incursion. We conceptualize the invention-induced change in this opportunity-to-exposure balance as a firm's *relative recombinant advantage*.

Our theory generates three predictions concerning how the relative recombinant advantage conferred by an invention shapes subsequent inventive activity. First, firms experiencing greater improvements in relative recombinant advantage should become more likely to undertake search incursions. Second, firms whose exposure increases proportionately more than their outward opportunities should become more susceptible to incursions by other firms. Third, an improvement in Firm A's overall relative recombinant advantage, accompanied by a deterioration in Firm B's, need not make Firm A particularly well positioned to undertake an incursion into Firm B's established inventive region. Firm A's improved opportunities may be concentrated toward Firm C, while Firm B's increased exposure may arise primarily from Firm D. The likelihood of an incursion by Firm A into Firm B's established inventive region therefore depends on the invention-induced changes in Firm A's overall opportunity-to-exposure balance, Firm B's corresponding balance, and the effort Firm A must expend to develop inventions in Firm B's established inventive region.

We tested these predictions by reconstructing how individual inventions changed the structure of technological knowledge and the costs of inventive search across firms. For each consequential invention, we reconstructed the patent-citation network over the preceding fifteen years and introduced the new invention together with its connections to prior technological knowledge. We conceptualized each firm as an actor searching through this network of patents, with search costs increasing as the distance between the firm's prior inventions and the knowledge being sought increased. We then employed a policy-iteration algorithm to estimate search costs before and after each invention, modeling the firm as a purposeful but imperfect "walker" through the network. Comparing the estimated search costs before and after the focal invention allowed us to estimate the resulting changes in firms' recombinant opportunity, exposure, and relative recombinant advantage. Figure 1 illustrats this procedure using an invention in high-efficiency infrared heating (US 4,727,854), introduced in 1988, and the technological knowledge positions of two firms, Shizuoka Seiki and Combustion Research, before and after the connections introduced by the invention.

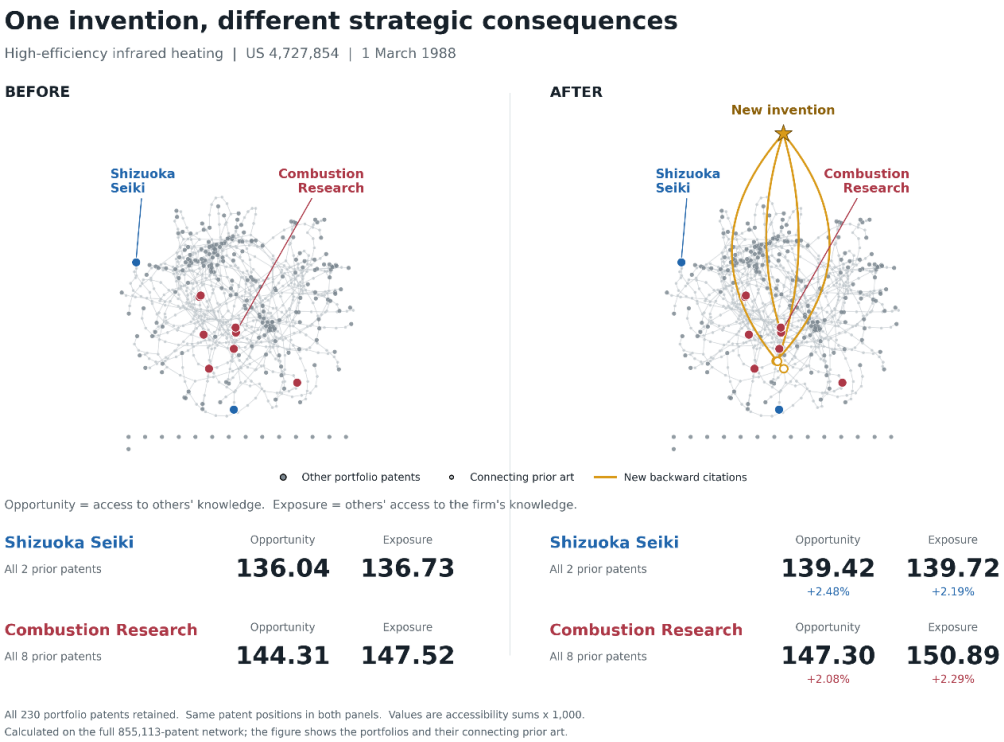


Figure 1: One invention, different strategic consequences: Shizuoka Seiki and Combustion Research

In this case, although the invention improved both firms' opportunities to pursue inventive activity beyond their established knowledge positions, its consequences for their relative recombinant advantages differed. Shizuoka Seiki experienced a 2.48 percent increase in recombinant opportunity against a 2.19 percent increase in exposure, while Combustion Research experienced a 2.08 percent increase in opportunity against a 2.29 percent increase in exposure. The same invention thus improved one firm's opportunity-to-exposure balance while deteriorating the other firm's, despite increasing technological accessibility for both. We extended this analysis across all consequential inventions, firms, and prospective incursive destinations in our sample, estimating how each invention changed both firm-level and destination-specific relative recombinant advantage.

Separately and in parallel, we used patent text and synthetic-patent constructions to identify which inventions qualified as search incursions. Finally, bringing these analyses together, we tested and found support for our theory by examining whether invention-induced changes in relative recombinant advantage predicted which firms undertook search incursions, which firms became susceptible to them, and which particular firm-to-firm incursions subsequently materialized.

1. Theory and Hypotheses

1.1 Search Incursion as a Construct

We conceptualize a firm's portfolio of inventions as a distribution of positions in a larger inventive space (Jaffe, 1986; Aharonson and Schilling, 2016). In this representation, each inventive artifact

can be seen as corresponding to a point, i.e an inventive position in a high-dimensional space whose dimensions capture the technological and functional attributes along which inventions vary (Fleming and Sorenson, 2004; Novelli, 2015). Proximity of points in this space can thus be viewed as indicating a similarity in the inventive content of the artifacts and distance indicates greater divergence (Arts, Cassiman, and Gomez, 2018). A firm's prior inventive activity in this conceptual inventive space typically appears as a concentration of inventive positions in some locations rather than others (Stuart and Podolny, 1996).

We define a search incursion as the act of a firm generating an invention that, in this conceptual inventive space (a) lies outside the set of locations in inventive space in which the focal firm has previously concentrated inventive effort and (b) lies within a location more firmly associated with the prior inventive positions of other firms. Given this definition, we can further clarify the construct by distinguishing it from two seemingly closely related accounts in the search literature. The first such distinction concerns the distinction between a search incursion and the concepts of local and more distant search (Cyert and March, 1963; Nelson and Winter, 1982; Stuart and Podolny, 1996). Applied to the invention's position in inventive space, the notion of local/distant search identifies one analytic dimension, in this case the invention's proximity to the focal firm's prior inventive positions.

In contrast, whether or not an invention can be deemed a search incursion cannot be fully specified by the invention's proximity to the firm's prior inventions alone but also requires consideration of the invention's similarity in its underlying content to the prior inventions of other firms. In sum, an invention that fulfils the definition of a search incursion may possibly be the result of exploration by the focal firm, or in other cases a search incursion may not particularly require extant exploratory effort but can simply be an exercise of previously non-actionable knowledge already available to the firm. The substantive correspondence across these two constructs may therefore most accurately therefore be viewed liberally as incidental, and most conservatively as correlational.

The analytical flavor of our prior discussion can also help clarify the relationship between search incursion and another closely comparable idea, that of boundary-spanning exploration. Rosenkopf and Nerkar (2001) classify exploration by firms in search according to whether the knowledge used in the process crosses organizational boundaries, technological boundaries, both, or neither. They distinguish between four resulting categories, those of local exploration, external boundary spanning exploration, internal boundary-spanning exploration, and radical exploration. In contrast to this focus on the source of knowledge, the classification of a search incursion concerns the invention and its functional relationship to the prior inventions of the focal firm and other firms.

To illustrate this distinction, a firm may for example draw on knowledge from outside its organizational boundaries that is technologically similar to knowledge already available within the firm (Rosenkopf and Nerkar, 2001). Such knowledge may then contribute to an invention that remains similar in inventive content to the firm's prior inventions. Establishing whether the inventive artifact can be deemed a search incursion therefore requires a separate assessment of the invention's similarity to the focal firm's prior inventions and to those of other firms. Consequently, the ideas of boundary-spanning exploration and search incursion might again incidentally overlap, while remaining orthogonal in their intended theoretical purpose.

1.2 Prior Research on Innovation Search and Technological Positioning

A complete theoretical account of search incursions requires an understanding of the circumstances under which firms might enter inventive regions populated by other firms and demands a treatment of the factors that make particular firms more likely to undertake (or be subjected to) such incursions. We draw on and depart from four related streams of research in the strategy literature: (a) innovation search, (b) recombinant invention, (c) knowledge networks and technological positioning, and (d) technological discontinuities, in the course of our theoretical development.

First, studies on innovation search have developed a substantial account of how attributes of knowledge components employed in the search process and the manner in which these knowledge components are combined shape the resulting inventive and product outcomes of the firm. Ahuja and Lampert (2001) examine search practices associated with the creation of breakthrough inventions, while Rosenkopf and Nerkar (2001) investigate how exploration across organizational and technological boundaries affects inventive impact. Katila and Ahuja (2002) distinguish the repeated use of existing knowledge from the incorporation of knowledge new to the firm and examine their effects on product introductions. Jung and Lee (2016) further characterize the originality of knowledge sought through search and its consequences for breakthrough invention.

Important studies have also examined the circumstances that merit a redirection of firms' inventive efforts. Greve and Taylor (2000) demonstrate how external innovations stimulate changes in organizational cognition and search, while Ahuja and Katila (2004) show how technological exhaustion and international expansion induce firms to pursue different technological paths. Ahuja, Lampert, and Tandon (2014) explain how firms' existing business configurations condition their technological responses to an external input-price shock.

Katila and Chen (2008) further establish how the timing of a firm's search relative to its competitors influences a firm's product innovation outcomes. These studies provide important explanations of the determinants, conduct, and consequences of firms' choices in their inventive search processes. We extend this understanding by departing to examine the particular destinations of inventive activity relative to other firms' prior inventions. Specifically, we investigate why the disequilibrating effects of an external invention might change the cost calculus confronting a firm and enable it to direct search efforts toward a region of the inventive space associated with other firms' prior inventive activity, and what factors determine which firm's region becomes the destination of such an incursion.

Second, related studies that start from a primary focus on recombination in relation to inventions form another foundational pillar that supports our present study in developing an understanding of how new technological possibilities can (a) catalyze search incursions and (b) generate the gradients of costs and benefits that dictate which firms might be benefited and thus impelled toward conducting search incursions, and which might instead be rendered susceptible to them. This literature conceptualizes invention as the combination of existing knowledge components into new configurations (Kogut and Zander, 1992; Fleming, 2001). The possibilities for recombination depend on the relationships among these components, including their technological interdependencies and the difficulty of identifying useful combinations (Fleming and Sorenson, 2001). Fleming and Sorenson (2004) explain how scientific knowledge can guide search across these possibilities, while Yayavaram and Ahuja (2008) establish how the structure of technological interdependencies influences inventive

usefulness and the malleability of knowledge bases.

A finding from this literature that forms a key antecedent to our present study is that new inventions can change the landscape of combinations for subsequent inventive activity. Bresnahan and Trajtenberg (1995) explain how general-purpose technologies create complementary opportunities for innovation, while Qu, Eggers, and Kumar (2026) examine how one prominent technology, artificial intelligence, enables recombinations across previously separated technological knowledge components. We break forward from this foundation by examining how external inventions can introduce new recombinations, and in doing so, induce firms to capitalize on this unlocking by conducting search incursions.

Third, research on knowledge networks provides the key basis for addressing another dimension in our present analysis. The relationships among technological knowledge elements influence the topology of search costs a firm encounters in its search process, i.e., which bodies of technological knowledge can be readily encountered and which require more extensive effort (Fleming and Sorenson, 2001, 2004). Carnabuci and Operti (2013), relatedly, demonstrate how the structure of intraorganizational knowledge networks conditions firms' recombinant capabilities, while Yayavaram and Ahuja (2008) establish the importance of relationships among knowledge components for subsequent invention.

These studies suggest that the opportunities available through recombinant search depend on the structure of existing technological associations, and we move outward from this reasoning toward an investigation of how a firm's position in an interfirm network of knowledge relationships, reconstructed from technological associations among firms' prior inventions, affects its future inventive choices about where to search.

Fourth, a complementary set of analyses examines firms' positions within the broader inventive landscape. Jaffe (1986) develops a framework for characterizing firms' technological positions, while Stuart and Podolny (1996) examine how local search shapes the evolution of technological capabilities and positions over time. Podolny, Stuart, and Hannan (1996) further establish the significance of technological niches and the relationships among firms occupying them. More recent research develops increasingly fine-grained representations of inventive space, including measures of firms' technological footprints and distances (Aharonson and Schilling, 2016) and their differentiation based on the semantic content of patents (Arts, Cassiman, and Hou, 2023). These studies establish how firms occupy identifiable regions of inventive space and how the relationships among these regions can be characterized, and this representation of firms forms the axle that is key to propel our theoretical development. In particular, it provides the basis for identifying when a firm's inventive activity departs from its prior inventive concentration and enters a region associated with another firm's prior inventions.

Finally, studies on general purpose technologies, creative destruction, and technological discontinuities focus on the disequilibrating effects of inventions and technologies (Dosi, 1982; Tushman and Anderson, 1986; Henderson and Clark, 1990; Bresnahan and Trajtenberg, 1995). While important strands of this literature focus on inventions and technologies that are at the right tail of impact in changing the macroeconomy and the fortunes of a large number of firms, our theory departs from this body of literature in two key ways. First, we depart toward a more general theory of inventions

and their disequilibrating effects, by taking a vast collection of consequential "less-than-GPTs", and understanding how this ocean of less conspicuous inventions continuously and incessantly redistributes the inventive opportunities underlying competitive advantage. Second, while several researchers have also examined heterogeneous firm-level responses to technological change (Tripsas, 1997; Rothaermel and Hill, 2005), our study can be viewed as an endeavor to progress technological change theory from categorical explanations of which types of firms benefit from technological changes toward a relational explanation of the particular firms that gain technological advantage over which particular other firms, following each particular invention.

2. Inventions, Disequilibrium, and Search Incursions

The value to a firm in conducting a search incursion in expectation may not be positive at all times, or even in a majority of cases, since firms possess existential incentives to defend "their" region of the inventive space with effectively placed entry/resource-position barriers. Therefore, firms might be more likely to conduct search incursions when a discontinuity in the environment improves the expected value of such an incursion, whether by creating new possibilities for inventive activity or by unsettling the barriers by which incumbents might retain advantage in a location in inventive space (Bain, 1956; Caves and Porter, 1977; Porter, 1980; Tushman and Anderson, 1986; Anderson and Tushman, 1990).

This environmental change might correspond to a regulatory, political, macroeconomic or technological development, among other possibilities. Our focus is on the technological case, i.e., where search incursions are enabled by an invention[2] that changes the search landscape firms face. A consequential invention can alter the feasible or economic frontier of technological action in that it lowers the cost of performing an activity, makes possible an activity that could not previously be performed, or both (Schumpeter, 1934; Rosenberg, 1982; Dosi, 1988; Bresnahan and Trajtenberg, 1995). When an invention has this effect, the topology of the broader search landscape of firms can be altered in a manner that might support search incursions through two possible mechanisms:

First, an external invention can change the search landscape by making feasible a recombination of knowledge components that previously could not be combined, paving the way for a search incursion (Fleming and Sorenson, 2001). To illustrate, let Firm A's prior inventive positions be concentrated around the bundle $\{o_A, a_1, a_2\}$, where that bundle denotes the knowledge components through which Firm A has historically generated inventive artifacts with a characteristic set of attributes and functionalities. Let Firm B's prior inventive positions be concentrated around $\{b_1, b_2, b_3\}$. A new invention, g_1, may then operate as a shared component that unlocks a newly feasible recombination such as $\{b_1, b_2, b_3, a_1, a_2, g_1\}$. Once this combination of knowledge elements becomes feasible to pursue, Firm B may then be able to produce an invention whose attributes and functionalities

[2] We focus on external inventions, i.e., inventions that enable search incursions but were created by neither the firm conducting the incursion nor the firm whose prior inventive region becomes its destination. This restriction is substantial: many, or perhaps most, incursions might be enabled by the firms' own inventions. We set aside this potentially predominant class to isolate how an invention external to both firms can alter their relative recombinant opportunities and disturb the competitive balance between them.

are broadly similar to those previously associated with the inventions in Firm A's prior inventive positions.

Second, an external invention can also change the search landscape by reconfiguring the entry barriers under whose protective shadow the "region" in the inventive space had been protected by the target firm. More formally, a firm's inventive advantage can depend on resources that competing firms would find difficult or costly to reproduce. Technological alternatives can diminish that advantage by substituting for the resources on which this advantage rests (Dierickx and Cool, 1989). To illustrate, suppose again that Firm A's prior inventive positions are concentrated around $\{o_A, a_1, a_2\}$. Suppose further that Firm A's advantage in that location of inventive space depends on preferential access to the knowledge or physical inputs corresponding to component o_A, control over the linkage between o_A and a_1 or a_2, or the production function through which that bundle is translated into a valuable inventive artifact. Under those conditions, Firm B may confront knowledge components and component combinations that are cognitively within range and technically feasible, yet have low search value because the existing production regime preserves Firm A's advantage.

Now, let g_1 denote an invention that enables Firm B to bypass the use of component o_A, reconfigures the linkage structure through which o_A connects to a_1 and a_2, or alters the production function through which that bundle yields a valuable inventive artifact (Henderson and Clark, 1990; Tushman and Anderson, 1986; Tripsas, 1997). If such changes occur, barriers rooted in the earlier production function may lose force (Bain, 1956; Caves and Porter, 1977; Porter, 1980; Afuah, 2000). Firms possessing expertise in g_1 may thus now be able to direct their search effort toward a region in the inventive space more firmly associated with Firm A's advantaged position.

3. Knowledge Positions, Relative Recombinant Advantage, and Search Incursions

In the preceding discussion, we discussed how the recombinatorial possibilities that an invention external to two firms generates might unlock opportunities for firms to conduct search incursions. However, our discussion above would appear to raise a serious query: if the disequilibrating effects of new inventions create possibilities for firms to conduct search incursions, what are the set of determinants that privilege one particular firm, say Firm A, over Firms B, C, and D to act on the opportunity? Conversely, what are the set of factors that determine if the disequilibrating invention leaves Firm D highly vulnerable to incursions from Firms A, B, and C?

Conducting a search incursion may be particularly attractive when, in addition to the weakening of entry barriers, Firm A can expect to maintain its advantage in its present activities and inventive trajectories while also faring favorably in an inventive and commercial contest against Firm B, in a domain in which Firm B possesses established expertise. Therefore, our concern extends beyond whether an invention unlocks new recombinations or weakens prior entry or resource-position barriers. We also examine whether (a) the invention recombines knowledge components in a manner that improves Firm A's costs of pursuing a search incursion and sustaining a strategic contest with Firm B, and (b) the invention changes the search landscape such that Firm B faces greater demands in continuing and defending its existing inventive trajectory, owing to increased exposure to incursions from multiple other firms. Where both occur, the invention creates particularly dire circumstances for Firm B. Our theory develops the conditions under which these changes arise.

Consider the overall inventive space to be represented as the space of all patents of firms, with directed edges indicating citations between patents (Stuart and Podolny, 1996; Podolny, Stuart, and Hannan, 1996). This overall inventive space need not necessarily be represented as a network, and alternative representations, including semantic embeddings, may also be suitable (Arts, Cassiman, and Hou, 2023). While the patent context forms a useful illustrative setting, our theory does not rely on any particular feature specific to the patent setting.

Figure 2 provides an illustration. In this figure, we highlight the inventions (nodes) of four firms associated with bicycle-related inventive activity: Bally Manufacturing (blue), Camp International (purple), Keiper Dynavit (green), and Schwinn Bicycle (red). Isolated patents are displayed at the bottom.

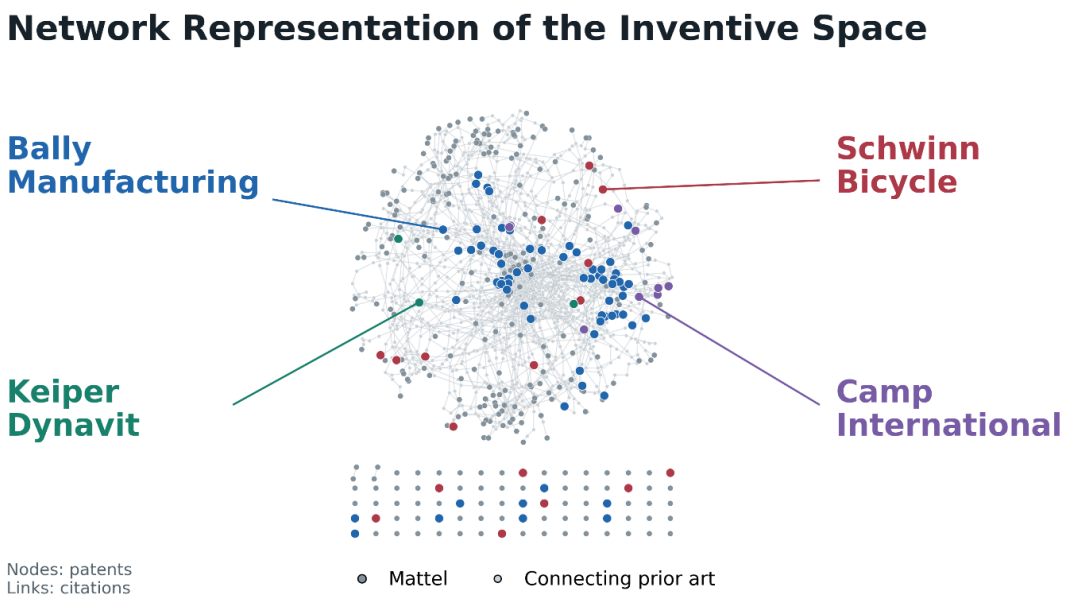


Figure 2: Firms' prior inventive positions in the technological knowledge network

Note. Nodes are patents; edges are citations. The prior inventions of Bally Manufacturing (blue), Camp International (purple), Keiper Dynavit (green), and Schwinn Bicycle (red) are highlighted; other patents appear in gray. Isolated patents are arrayed below the network. This panel precedes the external invention.

We then proceed from the foundational premise that firms have a predisposition to search locally, both as a consequence of rational considerations of the costs and expected returns to inventive effort, and because accumulated knowledge, experience, and cognitive limitations direct attention toward familiar technological possibilities (Cyert and March, 1963; Cohen and Levinthal, 1990; Stuart and Podolny, 1996). In our illustration, this would amount to the suggestion that the firm Keiper Dynavit might face a larger search cost in developing inventions with functionalities or attributes similar to those of Camp International's inventions, as opposed to those of Bally Manufacturing or Schwinn Bicycle.

This observation would imply that the present state of the inventive space denotes a corresponding matrix of search costs, describing the effort each firm would be expected to expend in searching toward the prior inventive regions of every other firm. Table 1 organizes these pre-invention search costs as a directed matrix for the four firms in our illustration:

$$C^{-} \equiv \left(c_{ij}^{-}\right)_{i,j\in\{B,C,K,S\}}$$

Table 1: Illustrative directed search-cost matrix before the external invention

	Bally	**Camp**	**Keiper**	**Schwinn**
Bally	–	c_{BC}^{-}	c_{BK}^{-}	c_{BS}^{-}
Camp	c_{CB}^{-}	–	c_{CK}^{-}	c_{CS}^{-}
Keiper	c_{KB}^{-}	c_{KC}^{-}	–	c_{KS}^{-}
Schwinn	c_{SB}^{-}	c_{SC}^{-}	c_{SK}^{-}	–

Here, each element c_{ij}^{-} denotes the expected search effort required for Firm *i*, beginning from its prior knowledge position, to encounter technological knowledge associated with Firm *j*'s prior inventive region. The rows therefore represent the firms undertaking search, while the columns represent their prospective inventive destinations. The diagonal elements are omitted because our concern is with search toward other firms' inventive regions.

For example, our preceding observation concerning Keiper Dynavit would correspond to a situation where:

$c_{KC}^{-} > c_{KB}^{-}$ and $c_{KC}^{-} > c_{KS}^{-}$

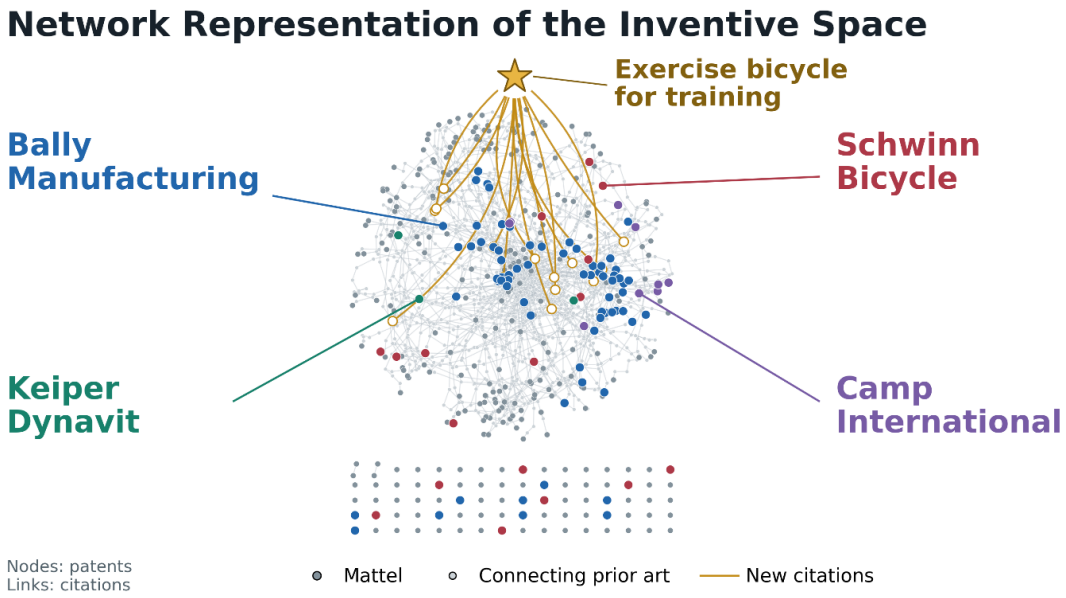


Figure 3: An external invention changes the overall cost structure of inventive search

Note. Prior patent positions and citation links are unchanged from Figure 2. The gold star marks Mock Bicycle for Exercise and Training Effects (US Patent 4,709,917; December 1, 1987); yellow links identify the invention's backward citations.

Now, we consider the events of 1987, when a new invention, titled "Mock Bicycle for Exercise and Training Effects" was patented by an independent inventor. Each invention can itself be imagined as the product of combination and recombination of prior knowledge elements (represented by yellow links in Figure 3).[3] The arrival of this invention brings with it a change in the above constructed matrix of search costs, because the underlying knowledge network has now changed. Simply put, the new invention has changed what counts as "local", to the four firms in our illustration.

With the external invention incorporated into the knowledge network, the corresponding directed search-cost matrix takes the revised form presented in Table 2:

$$C^g \equiv \left(c_{ij}^g\right)_{i,j\in\{B,C,K,S\}}$$

Table 2: Illustrative directed search-cost matrix after the external invention

	Bally	**Camp**	**Keiper**	**Schwinn**
Bally	–	c_{BC}^g	c_{BK}^g	c_{BS}^g
Camp	c_{CB}^g	–	c_{CK}^g	c_{CS}^g

[3] Recombination does not invariably establish a new technological connection, and inventions can alter inventive opportunities through mechanisms other than recombination. Our theory concerns the recombinant advantages arising from changes in technological relationships.

	Bally	**Camp**	**Keiper**	**Schwinn**
Keiper	c^g_{KB}	c^g_{KC}	–	c^g_{KS}
Schwinn	c^g_{SB}	c^g_{SC}	c^g_{SK}	–

The change induced by the external invention can therefore be represented as:

$$\Delta C^g = C^g - C^-$$

where each element of the matrix corresponds to a particular directed change in search costs:

$$\Delta c^g_{ij} = c^g_{ij} - c^-_{ij}$$

A negative value indicates that the external invention has reduced the expected cost of searching from Firm i toward Firm j's prior inventive region. These changes need not be equal across firms or destinations, nor between two opposing directions of search.

3.1 Recombinant Opportunity and Exposure

For any particular firm, the elements of each matrix would describe two distinct sets of search costs. The elements correspond to each firm's row in both matrices correspond to the costs the firm faces in searching toward other firms' inventive regions. The column elements correspond to the costs other firms face in searching toward the focal firm's inventive region, and the two matrices denote these costs before and after the external invention.

For Keiper Dynavit, for example, the elements c_{KB}, c_{KC}, and c_{KS} characterize its costs of searching toward Bally Manufacturing, Camp International, and Schwinn Bicycle, before the external invention. Conversely, c_{BK}, c_{CK}, and c_{SK} characterize the costs those firms face in searching toward Keiper Dynavit's inventive region, again before the external invention.

We refer to the costs a firm faces in searching toward other firms' inventive regions as determining its recombinant opportunity, and the costs other firms face in searching toward the focal firm's inventive region as determining its recombinant exposure.

More formally, for each state t before or after the external invention, recombinant opportunity is determined by the outward costs in a firm's row, while recombinant exposure is determined by the incoming costs in its column:

$$O^t_i = \mathcal{O}_i\left(\{c^t_{ij} : j \neq i\}\right)$$

$$E_i^t = \mathcal{E}_i\left(\{c_{ji}^t : j \neq i\}\right)$$

These expressions specify the directed relationships determining the two measures without imposing a particular aggregation rule on the network-based measures used in the empirical analysis.

Although recombinant opportunity and exposure can be conceptualized independently of any particular representation of technological knowledge, their measurement requires a representation of the relationships through which inventive search proceeds. In the empirical analysis that follows, we reconstruct networks of technological associations from patent citations and employ network algorithms and Monte Carlo simulations to construct measures of recombinant opportunity and exposure. We describe these procedures in detail in that section. For the purposes of the present discussion, higher values of opportunity correspond to lower outward search costs, while higher values of exposure correspond to lower incoming search costs.

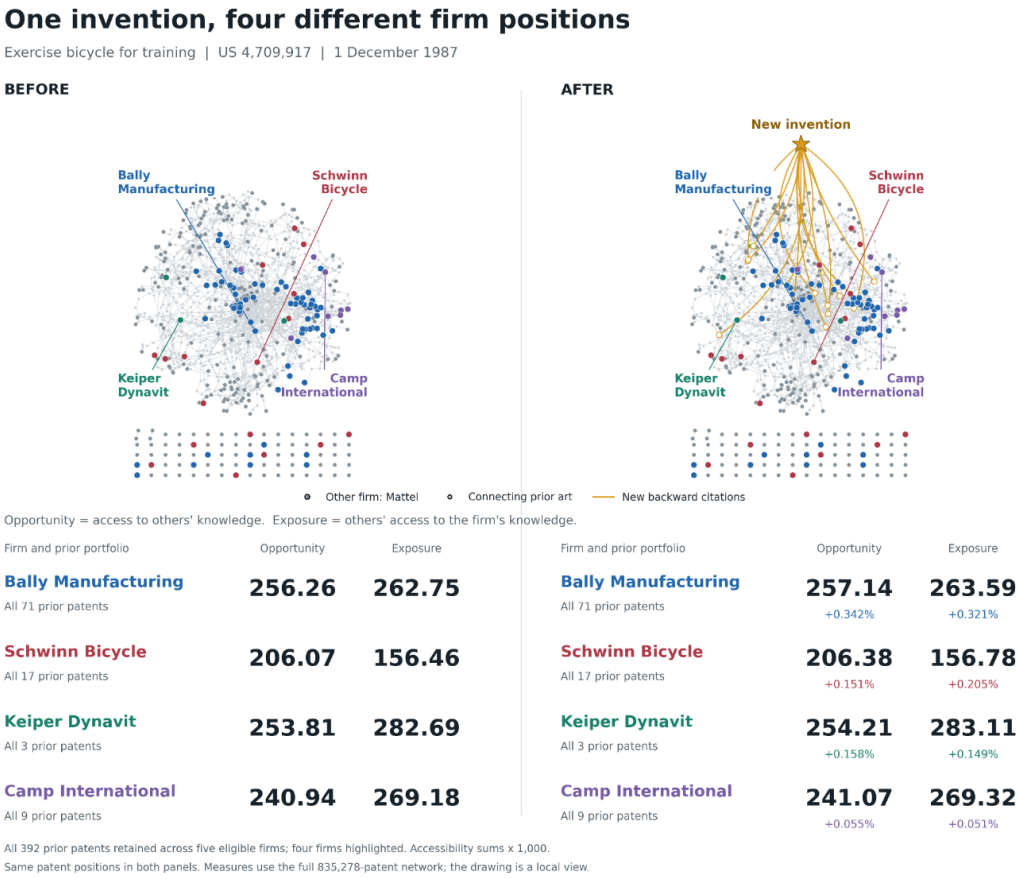


Figure 4: One invention, four different firm positions

Figure 4 brings the change in recombinant opportunity and exposure into view. The figure shows both the new technological connections introduced by the exercise-bicycle invention and the corresponding before-and-after accessibility measures for the four highlighted firms.

The numerical changes in Figure 4 reveal that the same external invention affects the four firms differently.

Table 3: Recombinant opportunity and exposure before and after the exercise-bicycle invention

Firm	Opportunity before → after	Exposure before → after
Bally Manufacturing	256.26 → 257.14 (+0.342%)	262.75 → 263.59 (+0.321%)
Schwinn Bicycle	206.07 → 206.38 (+0.151%)	156.46 → 156.78 (+0.205%)
Keiper Dynavit	253.81 → 254.21 (+0.158%)	282.69 → 283.11 (+0.149%)
Camp International	240.94 → 241.07 (+0.055%)	269.18 → 269.32 (+0.051%)

Table 3 shows that all four firms experienced improvements in both opportunity and exposure after the invention. Bally Manufacturing's opportunity grew proportionately faster than its exposure, while Schwinn Bicycle experienced the reverse. Keiper Dynavit and Camp International experienced much smaller differences between their proportional improvements in opportunity and exposure.

The importance of these differences follows from the resource demands associated with conducting a search incursion. As we discussed earlier, a firm seeking to establish an inventive position in another firm's region must undertake additional inventive activity while continuing to sustain and defend its established position.[4] Such competing claims upon inventive resources are consistent with the resource-allocation problem identified in organizational learning research (March, 1991; Nohria and Gulati, 1996), while the defense of established technological advantages can impose additional demands upon inventive resources (Dierickx and Cool, 1989; Ziedonis, 2004).

An improvement in recombinant opportunity has the effect of reducing the search effort required to pursue inventive activity beyond the firm's established region. On the other hand, increased recombinant exposure, meanwhile, can strengthen the prospective demands associated with defending that region against incursions by other firms. Consequently as we have seen in the case of the firms above, the same new incoming invention may relax the conditions governing a firm's outward inventive search while simultaneously increasing the claims upon resources required to preserve its established inventive position.

In Figure 5, we plot the invention-induced proportional increases in opportunity and exposure for the four firms in our example. The diagonal denotes equal increases in both quantities; for the firms corresponding to the points above it, the position indicates an improvement in the opportunity-to-exposure balance, while for the firms corresponding to points below it, this indicates a deterioration

[4] The resources required for incursions might also be mobilized through slack borrowed from future periods or resources made accessible through social networks. These possibilities could support alternative theories of search incursions. Our present account isolates the cost changes induced by an external invention.

in their overall position.

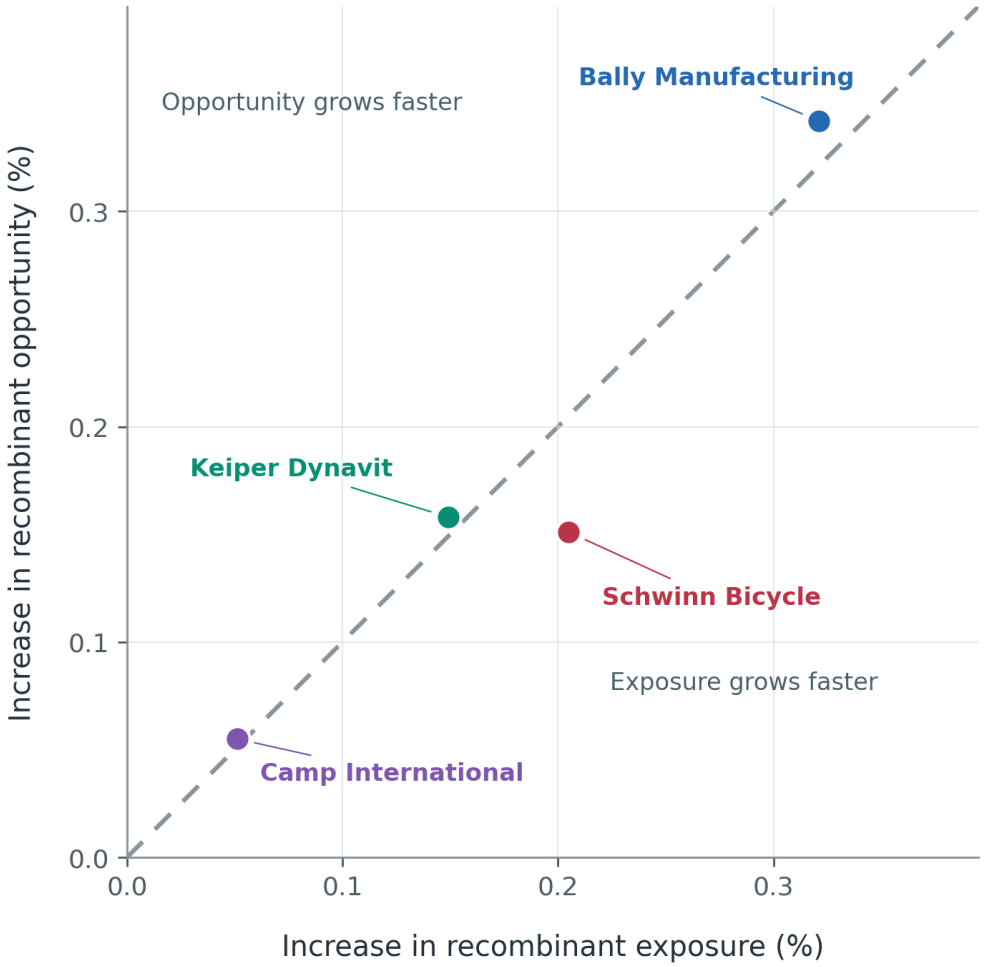


Figure 5: Changes in recombinant opportunity and exposure following US Patent 4,709,917

Note. The horizontal axis measures the percentage increase in exposure and the vertical axis the percentage increase in opportunity. The dashed diagonal denotes equal proportional changes; the observations are the four firms in Figure 4.

3.2 Relative Recombinant Advantage

Our argument therefore would be that the relative advantage conferred upon a firm by an external invention depends on how the invention changes the balance between these two considerations. We call this change a firm's relative recombinant advantage. For each particular firm, relative recombinant advantage compares the balance between its opportunities for outward inventive search and its exposure to incoming incursive activity by other firms before and after the external invention. In compact form:

$$\text{Relative recombinant advantage} \;=\; \frac{O_i^g / E_i^g}{O_i^- / E_i^-}$$

Here O^- and E^- denote the opportunity and exposure values before the external invention, while O^g and E^g denote their values after its introduction. An point of note here is that this comparison concerns the change induced by the external invention, rather than the absolute levels of opportunity and exposure prevailing before its appearance.

Writing the invention-induced proportional change in this balance explicitly:

$$\Delta R_i^g = \frac{O_i^g / E_i^g}{O_i^- / E_i^-} - 1$$

A positive value denotes an improvement in the opportunity-to-exposure balance; a negative value denotes a deterioration. This is the proportional-change form of the preceding comparison.

Returning to Figure 4, Bally Manufacturing's opportunity increased by 0.342 percent while its exposure increased by 0.321 percent, producing an improvement in its opportunity-to-exposure balance. Schwinn Bicycle, by contrast, experienced an increase of 0.151 percent in opportunity and 0.205 percent in exposure, resulting in a deterioration in that balance.

The first implication of our argument then concerns which firms become more likely to undertake search incursions following an external invention. Among firms for which the invention creates technologically feasible and potentially valuable opportunities, an improvement in relative recombinant advantage should indicate a favorable change in the conditions governing incursive search.[5] This means that the firm can pursue external inventive regions at reduced search cost without confronting a commensurate increase in its exposure to incursions by others. Other relevant determinants of inventive activity held constant, the invention can be seen to have consequently improved the attractiveness of allocating inventive resources toward an incursion.

We can therefore expect the likelihood of a firm conducting a search incursion to increase with the improvement in relative recombinant advantage conferred upon the firm by the external invention.

> **Hypothesis 1:** Among firms for which an external invention creates feasible and valuable inventive opportunities, the likelihood of undertaking a search incursion increases with the invention-induced improvement in the firm's opportunity-to-exposure balance.

The corresponding argument can be further extended to hypothesize on which firms might become particularly susceptible to search incursions following an external invention. A deterioration in a firm's opportunity to exposure balance indicates that the invention has created comparatively more

[5] The applicability of an invention to a firm need not imply that it improves the firm's access to another firm's inventive region. A prospective target need not itself build upon the external invention.

favorable conditions for other firms to pursue inventive activity within the focal firm's established region than for the focal firm to extend its own inventive activity into their regions.

We therefore expect a firm's susceptibility to search incursions to increase with the extent to which the invention improves its exposure relative to its opportunity.

> **Hypothesis 2:** Among firms whose prior inventive regions are potential destinations, the likelihood of becoming the target of a search incursion increases with the invention-induced improvement in the firm's exposure-to-opportunity balance.

3.3 Relative Recombinant Advantage and Incursive Destinations

The preceding hypotheses concern how an external invention changes firms' overall positions for undertaking and receiving search incursions. These positions, however, aggregate search relationships with many different firms. An improvement in Firm A's relative recombinant advantage need not make Firm B a more favorable destination, since the improvement may arise principally from reduced costs of reaching other firms. Likewise, a deterioration in Firm B's position may reflect greater exposure to firms other than A. Explaining the direction of a particular incursion therefore requires examining the relationship between the prospective origin and destination, alongside their respective changes in relative recombinant advantage.

We argue that three invention-induced changes jointly create particularly favorable circumstances for an incursion from Firm A into Firm B's established inventive region. First, the invention improves A's opportunity-to-exposure balance, strengthening its overall position for undertaking incursions. Second, the invention deteriorates B's opportunity-to-exposure balance, leaving its established inventive region relatively more exposed to entry by other firms. Third, the invention reduces the cost of searching specifically from A toward B's region. In terms of our earlier measures, these conditions are:

$$\Delta R_A^g > 0, \qquad \Delta R_B^g < 0, \qquad \Delta c_{AB}^g < 0.$$

Each condition contributes a distinct consideration. A reduction in A's cost of reaching B may occur without a corresponding improvement in A's overall opportunity-to-exposure balance. A may consequently gain access to B while facing equally substantial improvements in other firms' access to its own established region. Similarly, A's position may improve without B becoming relatively more exposed to incoming search. Most importantly, both firms' overall positions may change in the predicted directions without any reduction in the particular cost of searching from A toward B.

Consider, for example, an invention that principally improves A's access to Firms C and D while increasing B's exposure to incursions from those same firms. A's relative recombinant advantage could improve and B's deteriorate, even though the cost of searching from A toward B remained unchanged. The invention would have altered both firms' overall positions without improving the specific directed relationship connecting them. The third condition therefore identifies which

prospective destinations become more favorable to a particular incursor, beyond the changes in firms' aggregate positions.

We capture these considerations through destination-specific recombinant advantage. For a prospective incursion from firm i toward firm j, we define:

$$S_{ij}^{s} = \frac{1}{1 + c_{ij}^{s}} \frac{R_i^s}{R_j^s},$$

where c_{ij}^{s} denotes the directed cost of searching from i toward j in network state s, and R_i^s and R_j^s denote the firms' respective opportunity-to-exposure balances. The first term captures the specific search relationship, while the ratio of balances incorporates the origin's overall position relative to the prospective destination's position.

We measure the change conferred by the external invention as:

$$\Delta S_{ij}^{g} = \frac{S_{ij}^{g}}{S_{ij}^{-}} - 1.$$

Holding the remaining components constant, destination-specific recombinant advantage increases when the invention reduces the directed search cost, improves the origin's opportunity-to-exposure balance, or deteriorates the destination's balance. The three favorable changes reinforce one another when they occur together. Their conjunction therefore represents particularly favorable circumstances for a directed incursion, although an improvement in the combined position may also arise from favorable changes in fewer than all three components.

For a given prospective incursor and external invention, the origin's change in relative recombinant advantage is common across its possible destinations. The differences among those destinations therefore depend on their own changes in opportunity-to-exposure balance and on the changes in the costs of searching toward them. Figure 6 separates these destination-specific relationships: two targets can differ both in the cost of reaching their knowledge and in the balance of opportunity and exposure at their respective positions.

We consequently expect an observed incursor to be more likely to enter a particular firm's established inventive region when the focal invention improves its destination-specific recombinant advantage toward that firm, conditional on the existence of technologically feasible and potentially valuable inventive opportunities.

> **Hypothesis 3:** Following an external invention, for a firm undertaking a search incursion, the likelihood of entering a particular other firm's established inventive region increases with the invention-induced improvement in destination-specific recombinant advantage toward that firm.

4. Empirical Design

To test our hypotheses, we first identified influential inventions and subsequent search incursions, then examined how estimated search costs changed following each focal invention. We used U.S. utility patents to establish the invention universe (Griliches, 1990; Ahuja & Lampert, 2001), patent text to identify incursions (Arts et al., 2018; Kelly et al., 2021), and patent citation relationships to estimate directed search costs and examine their association with firms' subsequent incursion activity.

4.1 Identifying Influential Inventions

To identify inventions capable of reshaping the costs of inventive search and creating opportunities for search incursions, we began with approximately 7.3 million U.S. utility patents. Our theory directed attention to inventions exhibiting three characteristics. First, an invention should support subsequent inventive activity across multiple firms and technological domains, creating opportunities for firms with different established knowledge bases to pursue new inventive regions. Second, it should draw upon technologically disparate antecedents, providing connections through which previously separated knowledge domains may become accessible to one another. Third, it should provide a distinct basis for subsequent inventive activity, since inventions that primarily extend existing technological trajectories offer less scope for creating new recombinant opportunities. Together, these characteristics identify inventions whose influence may extend beyond their immediate technological setting and alter the relationships among firms' established inventive positions.

We operationalized these characteristics by adapting the three dimensions of the measurement framework developed by Rathje and Katila (2021). To capture the diversity of subsequent applications, we measured downstream technological breadth as one minus the Herfindahl concentration index across the three-digit USPC classes of patents citing the candidate invention within fifteen years of its grant. We also required citations from at least ten distinct firms, establishing that its subsequent influence extended across multiple organizations. To capture the diversity of technological antecedents, we measured upstream technological breadth using the same concentration measure across the technological classes of patents cited by the candidate invention. Finally, to assess whether subsequent inventors treated the invention as a distinct technological reference, we measured distinct downstream use as the proportion of subsequent citing patents that cited the candidate invention without also citing any of its own cited predecessors. We assigned zero to missing component values, standardized the three measures, and averaged them to obtain a composite score for ranking candidate inventions.

Although these measures captured breadth of technological antecedents, diversity of subsequent applications, and distinctiveness within downstream citation patterns, a high composite score did not necessarily establish that an invention provided a substantive new basis for subsequent inventive activity. We therefore supplemented the patent-based ranking with an assessment using a large language model (LLM). The LLM examined patent descriptions and associated citation evidence to assess whether candidate inventions supported distinct derivative inventive activity rather than principally reflecting overlapping or narrowly bounded technological variants. We evaluated 11,043 candidates and ordered them by their composite scores. The sequence of classifications in Figure 7 shows affirmative decisions concentrated among higher-ranked inventions, with negative decisions

becoming substantially more frequent beyond the first 6,000 candidates. We therefore retained the first 6,000 reviewed inventions as the screening population, within which 5,548 received affirmative LLM classifications.

We then restricted these affirmative candidates to inventions granted between 1985 and 2009, yielding 5,123 focal influential inventions. The upper boundary provided a complete fifteen-year period for observing subsequent citations through 2024, while the lower boundary allowed us to observe historical patenting activity preceding the focal inventions, subject to the available coverage of earlier patent records. The selected inventions were subsequently cited by 24,606 distinct corporate firms, yielding 90,930 focal-invention–firm observations, with an average of 17.75 citing firms per invention. Table 4 situates this sample across grant cohorts and technological sectors, while Table 5 gives concrete examples of inventions retained and excluded by the screening procedure. The backward citations of our earlier exercise-bicycle invention example provide a further illustration of the upstream breadth measured in our selection process. In Figure 8, we show how the invention distributes its fourteen historical-network references across eight primary USPC classes.

4.2 Identifying Search Incursions

In our theory, a search incursion occurs when a firm produces an invention outside its established inventive region and within a region more firmly associated with another firm's prior inventions. Identifying such incursions requires comparing the subsequent invention with the historical knowledge of both its originating firm and firms whose established inventive regions constitute potential destinations. For each focal influential invention g, we therefore identified firms whose patents cited g within fifteen years of its grant. Their citations provided observable evidence of a shared technological interest in g, making these firms relevant to our proposed mechanisms of recombination feasibility and barrier reconfiguration, through which a common invention may enable entry into technological regions previously developed by others.

Identifying this population required a methodological compromise. Comparing every subsequent citing patent against the historical knowledge of the entire patenting population would require potentially billions of evaluations, including firms with no observed connection to the focal invention. Identifying relevant firms prospectively through a predictive model would introduce a separate classification problem, whose errors would carry into our measurement of incursions. We therefore relied on observed citations during the fifteen-year follow-up to identify firms with a demonstrated connection to g. This necessarily introduces look-ahead information, since the eventual citing population cannot be established at the time of the focal invention's grant. We accepted this limitation as the consequence of identifying relevant firms through observed inventive activity rather than estimated technological relevance. Accordingly, our measure captures incursions among firms subsequently observed citing the same focal invention.

To establish the prior inventive knowledge against which subsequent patents would be evaluated, we drew exclusively on patent histories preceding the focal invention. We retained identifiable corporate citing firms and excluded universities, government and military organizations, nonprofits, unresolved organizations, and identified self-citations. For each firm, we selected up to twenty textually relevant utility patents with usable abstracts from its portfolio during the fifteen years

preceding g. These patents represented the firm's historical knowledge for the textual comparisons. We also excluded prior patents sharing recorded ownership with the subsequent citing patent from external-firm comparisons.

Having established these historical portfolios, we assessed whether each subsequent citing patent q, produced by firm i, more closely resembled the originating firm's prior knowledge or that of other participating firms. Because all participating firms cited g, their subsequent patents could share terminology associated with the focal invention itself. We therefore constructed two word-frequency distributions that assigned identical weight to g's abstract, allowing the comparison to depend on the historical patent knowledge incorporated alongside it. The first distribution assigned half its probability weight to g's abstract and half to firm i's selected prior patent abstracts. The second assigned the same weight to g, with the remaining half drawn from the historical patent abstracts of other eligible firms citing g. To construct the external distribution, we first pooled and normalized word counts within each eligible external firm's historical patent abstracts and then averaged these firm-specific distributions with equal weight, preventing larger portfolios from dominating the comparison.

Let P_g denote the normalized word distribution of the focal invention's abstract, P_{ig} the corresponding distribution of the originating firm's selected historical patent abstracts, and $\overline{P}^{(q)}_{-i,g}$ the equally weighted distribution across eligible external firms after excluding patents sharing ownership with q.

We defined the two distributions as follows:

$$M^{\text{own}}_{igq} = 0.5P_g + 0.5P_{ig},$$
$$M^{\text{external}}_{igq} = 0.5P_g + 0.5\overline{P}^{(q)}_{-i,g}.$$

To determine which distribution more closely reflected the observed abstract of q, we independently generated 1,000 simulated texts from each distribution by sampling words with replacement and matching each text's length to that abstract. We then calculated cosine similarity between each simulated text and q using raw word-count vectors. For each paired comparison, we recorded whether the external distribution generated a strictly closer textual match than the originating firm's distribution, treating ties as non-wins. We defined the search-incursion score as:

$$\widehat{p}_{igq} = \frac{1}{1000}\sum_{b=1}^{1000} \mathbf{1}\left[\cos\left(q, \tilde{q}_b^{external}\right) > \cos\left(q, \tilde{q}_b^{own}\right)\right].$$

A score of 0.75, for example, indicated that the external distribution generated a closer textual match in 750 of the 1,000 comparisons. Higher scores indicated stronger relative alignment with the pooled historical patent knowledge of other participating firms, holding the focal invention's textual weight constant. The score represents a simulated comparison frequency rather than a calibrated probability that an incursion occurred.

Each observation comprised a focal invention, a corporate citing firm, and a subsequent citing patent. Of the 263,680 observations, 139,644 had sufficient information for scoring. The remaining 124,036 were unscored, principally because the originating firm had no selected prior patents (122,068 observations). We treated these observations as unavailable rather than classifying them as non-incursions. Among scored observations, the median search-incursion score was 0.498, while the 90th, 95th, and 99th percentiles were 0.617, 0.667, and 0.802, respectively. To identify unusually strong alignment with external historical knowledge, we examined the upper tail of the pooled score distribution. An endpoint-chord criterion and a two-line fitting criterion identified elbows at the 96th and 98th percentiles, respectively; their midpoint selected the 97th-percentile score of 0.709 as the classification threshold. Figure 9 locates this cutoff in the score distribution and its upper tail. Applying this inclusive cutoff identified 4,212 search-incursion events, representing 3.02 percent of scored observations, involving 3,888 distinct patents across 1,727 focal inventions. Table 6 traces the corresponding sample counts from the corporate adoption universe through scoring, incursion identification, and network retention.

The pooled comparison established whether a subsequent invention exhibited unusually strong alignment with external historical knowledge, but it did not identify the particular firm whose prior inventive region the invention most closely approached. We therefore determined destinations separately, preventing the strongest individual match among candidate firms from determining the initial incursion classification. For each qualifying observation, we repeated the comparison against each eligible external firm's historical patent distribution, retaining the same focal-invention contribution and originating-firm comparison. We attributed the incursion to the external firm with the highest target-specific frequency of closer textual matches, resolving exact ties using the lowest organization identifier. This procedure yielded 2,773 distinct directed focal-invention–origin–destination relationships, involving 1,346 originating firms and 1,197 destination firms. Multiple qualifying observations involving the same focal invention and ordered firm pair contributed one directed relationship. Figure 10 makes the threshold comparison concrete with two patents citing the same focal invention, one classified as an incursion and the other below the cutoff.

Finally, to test our measure, we also examined how the identified incursion roles differed across previously established measures of firms' prior inventive activity from the strategy literature. Table 7 compares these characteristics across the retained network sample, separating firms observed exclusively as incursion origins, exclusively as destinations, in both roles, or in neither observed role. Origin-only firms had smaller prior patent portfolios than destination-only firms (853 versus 2,519 patents), shorter observed patenting histories (6.8 versus 12.2 years), and lower patenting momentum (0.38 versus 0.89). Their historical knowledge bases exhibited lower technological breadth (0.35 versus 0.73), alongside greater prior search scope (0.86 versus 0.74) and greater semantic distance from the focal invention (0.92 versus 0.81). Firms with neither observed role had larger prior patent stocks (3,455 patents), greater technological breadth (0.79), and lower prior search scope (0.73) than origin-only firms. Prior-patent originality differed comparatively little across origin-only, destination-only, and neither-role firms (0.42, 0.40, and 0.41, respectively).

4.3 Measuring Directed Search Costs and Recombinant Advantage

Having identified search incursions and their destinations, we next examined how focal inventions changed the technological conditions governing search between firms' established inventive regions. Our theory locates these changes in the directed costs of reaching another firm's prior knowledge and in the resulting distribution of recombinant opportunity and exposure. We therefore estimated how each focal invention altered the costs of searching between eligible firms, using the network of technological knowledge preceding its appearance. We restricted this analysis to the 1,727 focal inventions with at least one observed search incursion, retaining all eligible citing firms associated with each invention, including those without an observed incursion. This allowed us to examine differences among prospective origins and destinations within the invention-specific populations in which incursions occurred.

The relevant search costs are directional. The effort required for firm i to reach firm j's established inventive region may differ from the effort required for j to reach i's region. Moreover, a focal invention may reduce these costs by different amounts across firms and directions, altering their relative opportunities for inventive search. To capture these differences, we constructed a matrix of directed search costs for each focal invention and estimated how its entries changed when the invention entered the existing technological network.

Estimating these costs required establishing the historical inventive positions between which firms could search. We represented inventive knowledge as a network in which patents constituted nodes and their citations established technological connections. For each focal invention g, we reconstructed the network of eligible patents granted during the preceding fifteen years and their citation relationships. We then selected one representative utility patent from each eligible firm's historical portfolio to serve as its starting position and as a destination for other firms' search. We selected the patent with the shortest undirected citation-hop distance to g, measured through the focal invention's backward citations, resolving ties using BM25 textual similarity to g's abstract and then patent identifiers. We held these representative patents fixed across the subsequent calculations so that changes in search costs referred to the same originating and destination positions. Although our comparisons concerned firms associated with the same focal invention, search could traverse the entire eligible historical patent network, including patents belonging to other organizations. The representative patent therefore served a different purpose from the up-to-twenty-patent portfolios used in the preceding subsection: those portfolios characterized historical knowledge for identifying incursions, whereas the representative established a fixed location for calculating network search costs.

Given these starting positions and destinations, we modeled search as purposeful navigation through the patent network under uncertainty. A searcher could select a neighboring patent connected through a citation relationship at a cost of one unit. The intended move succeeded with probability 0.90; with probability 0.10, movement instead followed a fixed background distribution over the patent's original neighbors, with a self-loop for isolated patents. The searcher could also make an exploratory jump at a cost of ten units, landing uniformly at random on a patent in the original network. This option permitted access to otherwise disconnected regions, at a higher cost than local movement. The searcher selected among these actions to minimize the expected cumulative cost of reaching a

specified destination.

Let C_{ij}^{s} denote the minimum expected search cost from firm i's representative patent to firm j's representative patent in network state s, with $s = -$ representing the historical network before the focal invention and $s = g$ the network after its introduction.

We calculated the minimum expected search cost as:

$$C_{ij}^{s} = \min_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{\tau_j - 1} c_t \;\middle|\; X_0 = r_i, \mathcal{G}^s \right].$$

Here, π denotes a search policy, c_t the cost of the action selected at step t, r_i firm i's representative patent, and τ_j the first arrival at firm j's representative patent. We solved the optimization using policy iteration and sparse linear algebra, repeatedly evaluating and improving search policies until convergence. The resulting costs incorporate the consequences of navigation errors and exploratory moves as well as the available citation connections. They therefore differ from the citation-hop distances used to select firms' representative patents. Furthermore, although citation associations were traversable in either direction, differences in the stochastic navigation process could produce unequal costs in opposite directions. These quantities measure expected search costs in model units rather than observed monetary expenditures.

To isolate the change associated with each focal invention, we first estimated directed search costs in the historical patent network. We then introduced g and its eligible backward-citation connections and recalculated the costs, preserving the original patents, existing connections, representative positions, navigation-error distributions, and exploratory-jump population. No subsequent patents or forward citations entered the augmented network. We defined the invention-induced change as:

$$\Delta C_{ij}^{g} = C_{ij}^{g} - C_{ij}^{-}.$$

A negative value indicates that the focal invention reduced the expected cost of searching from firm i toward firm j's established inventive position. Because the augmented network preserved all previously available search actions, the estimated minimum costs could decrease or remain unchanged. Their reductions, however, could differ considerably across firms and directions. Thus, even when the invention made particular regions easier to reach, it could improve some firms' relative search positions more than others.

At the firm level, these directional changes have two implications. Recombinant opportunity concerns a firm's ability to reach other firms' established inventive regions, while recombinant exposure concerns other firms' ability to reach its own region. To measure both dimensions, we transformed each directed cost using $1/(1 + C)$, which assigns greater weight to less costly search relationships. For each focal invention, we held the set of eligible comparison firms, F_g, fixed and calculated:

$$O_i^s = \sum_{\substack{j \in F_g \\ j \neq i}} \frac{1}{1 + C_{ij}^s},$$
$$E_i^s = \sum_{\substack{j \in F_g \\ j \neq i}} \frac{1}{1 + C_{ji}^s}.$$

Recombinant opportunity therefore summarizes the costs of outward search from firm i, while recombinant exposure summarizes the costs of search directed toward it. Both quantities refer to the same eligible firms and are calculated in each network state.

Because a focal invention may improve a firm's outward search opportunities while also making its established region easier for others to enter, our theory concerns the balance between these two changes. We therefore calculated each firm's opportunity-to-exposure balance and its proportional change following the focal invention:

$$R_i^s = \frac{O_i^s}{E_i^s}, \qquad \Delta R_i^g = \frac{R_i^g}{R_i^-} - 1.$$

The change ΔR_i^g measures the relative recombinant advantage conferred by the focal invention. A positive value indicates that recombinant opportunity increased proportionally more than exposure, while a negative value indicates that exposure increased proportionally more than opportunity. We used this change to examine whether firms gaining relative recombinant advantage were more likely to undertake search incursions, as predicted by Hypothesis 1, and whether firms experiencing a deterioration were more likely to become incursion destinations, as predicted by Hypothesis 2.

The first two hypotheses concern firms' overall positions, whereas our third hypothesis concerns the direction of particular incursions. An improvement in firm i's relative recombinant advantage may arise from reduced search costs toward firms other than j, leaving the conditions governing a potential incursion from i into j's region comparatively unchanged. Identifying favorable destinations therefore requires considering the directed relationship between the two firms alongside their respective opportunity-to-exposure balances. We captured these conditions through destination-specific recombinant advantage, calculated as:

$$S_{ij}^s = \frac{1}{1 + C_{ij}^s} \frac{R_i^s}{R_j^s}, \qquad \Delta S_{ij}^g = \frac{S_{ij}^g}{S_{ij}^-} - 1.$$

Holding the other components constant, destination-specific recombinant advantage increases when the search cost from i toward j falls, the originating firm's opportunity-to-exposure balance improves, or the destination firm's balance deteriorates. The proportional change ΔS_{ij}^g measures how the focal invention alters this combined position for a particular ordered firm pair. We used it to operationalize

the joint conditions governing destination-specific incursions under Hypothesis 3. Accordingly, the three empirical specifications examine the invention-induced change in the prospective origin's relative recombinant advantage, the corresponding change for the prospective destination, and the change in destination-specific recombinant advantage for the ordered firm pair.

The full network sample comprised 21,646 firm–invention observations and 363,390 directed firm pairs across the 1,727 focal inventions. We calculated firms' network positions using their complete eligible comparison sets before applying the subsequent sample restrictions. The retained descriptive sample comprised 21,102 firm–invention observations, of which 20,725 had known origin status, and 349,353 directed firm-pair observations with known origin status. Table 8 reports the distributions of recombinant opportunity, exposure, and their invention-induced changes, including destination-specific recombinant advantage. Figure 11 connects these measures to the underlying patent-network intervention: an external invention creates new citation links, which alter directed search costs and the outward and incoming accessibility of firms' historical inventive positions.

4.4 Variable Definitions and Model Specification

4.4.1 Dependent Variables

The search-incursion procedure identified which firms produced qualifying incursions and which firms' historical inventive regions those incursions entered. We used these observations to examine three outcomes corresponding to our hypotheses: whether a firm undertook an incursion, whether its established inventive region became a destination, and which particular destination an observed incursor entered. The first two outcomes were measured for each firm associated with a focal invention; the third was measured for each eligible ordered firm pair around that invention.

We defined incursion origin (Y_{ig}^{O}) as an indicator equal to one if firm i produced at least one qualifying search incursion associated with focal invention g. A single observed incursion was sufficient to establish a positive outcome. Establishing a negative outcome, however, required complete information: we assigned zero only when every relevant citing-patent observation had been scored and none qualified as an incursion. Otherwise, origin status remained unavailable.

We defined incursion target (Y_{jg}^{T}) as an indicator equal to one if firm j was the attributed destination of at least one observed incursion associated with g. We assigned zero when no incoming incursion was attributed to the firm. This outcome captured observed entry into a firm's established inventive region without implying economic harm. Because destination status depended on incoming attributions, it could be observed even when the firm's own origin status was unavailable.

These two indicators identified firms' roles but did not establish which particular firm pairs experienced incursions. To examine their direction, we defined destination selection (Y_{ijg}^{D}) for each eligible destination j of a firm i already observed undertaking an incursion around g. The indicator equaled one if at least one qualifying incursion from i was attributed to j, and zero if no such attribution was observed. We compared every eligible destination separately. Consequently, an originating firm could have multiple positive destination outcomes, and the analysis concerned the incidence of incursions into particular regions rather than a mutually exclusive choice of one destination.

4.4.2 Independent Variables

Our independent variables corresponded to the invention-induced changes in recombinant advantage developed in the preceding subsection. For the first two hypotheses, we used relative recombinant advantage (ΔR_i^g), the proportional change in a firm's outward recombinant opportunity relative to its incoming exposure following the focal invention. We expected larger improvements to increase the likelihood that a firm undertook an incursion (Hypothesis 1) and reduce the likelihood that its own established inventive region became a destination (Hypothesis 2). The same measure therefore entered both firm-level specifications, with opposite predicted associations.

For Hypothesis 3, we used destination-specific recombinant advantage (ΔS_{ij}^g). This measure incorporated the invention-induced change in the directed search relationship from prospective origin i toward destination j, together with the changes in both firms' opportunity-to-exposure balances. We expected larger improvements to increase the likelihood that an observed incursor entered the corresponding destination's inventive region.

4.4.3 Control Variables

Firms' established inventive capabilities and search histories could influence their incursion activity independently of the changes in recombinant advantage associated with the focal invention. We therefore included sixteen firm-level controls describing their inventive resources, prior knowledge, search practices, and network positions. The underlying patent histories used to construct these controls preceded the focal invention's grant.

We first accounted for differences in the scale and development of firms' inventive activity. Prior patent stock measured the logarithm of one plus the number of previously recorded firm patents, including all patent types. Observed patenting age measured the logarithm of one plus the years elapsed since the firm's first recorded patent, capturing its observed patenting history rather than its age since incorporation. Patenting momentum measured changes in recent inventive activity as the difference between logged annual patenting rates in the five years immediately preceding the focal invention and the preceding five-year period.

We next accounted for the composition and organization of firms' historical knowledge. Prior-patent originality measured the average diversity of technological classes among the backward citations of each firm's selected historical patents, using one minus the Herfindahl concentration index. Knowledge-base connectedness measured the proportion of patent pairs connected through citation paths within the firm's preceding fifteen-year utility-patent portfolio. Since connectedness was undefined for a portfolio containing only one patent, we included a separate one-patent portfolio indicator. Technological breadth measured the diversity of primary three-digit USPC classes represented in the firm's historical utility-patent portfolio, again using one minus the Herfindahl concentration index.

The relationship between a firm's prior knowledge and the focal invention could also affect its subsequent inventive activity. We therefore included semantic distance to the focal invention, measured as one minus the maximum TF-IDF cosine similarity between the focal invention and the firm's selected historical patent abstracts. This textual measure differed from the citation-network distance used to select each firm's representative patent. We also controlled for selected prior portfolio size, measured as the logarithm of one plus the number of usable patents retained for the

historical textual comparison, up to twenty.

To capture established search practices, we included prior search depth, which measured earlier reuse of cited knowledge during the five years preceding each selected historical patent, and prior search scope, measured as the proportion of references absent from both the firm's earlier patented knowledge and its preceding five-year citation history. We also controlled for external-citation share, the proportion of scorable references assigned to other organizations, treating coowned references as internal. Two further measures characterized the historical knowledge sources available to the firm: knowledge-base vintage, the mean age of its selected prior patents at the focal invention's grant, and external-source nonredundancy, measured as one minus citation-network density among the external organizations referenced by those patents.

Finally, firms could begin with substantially different positions in the historical patent network. We therefore included baseline network scale, $\log\left(O_i^- + E_i^-\right)$, and baseline opportunity-to-exposure balance, $\log\left(R_i^-\right)$. These measures accounted for the firm's initial outward and inward search position when estimating associations with the changes induced by the focal invention.

The destination-selection analysis additionally accounted for relationships that already existed between prospective origins and destinations. Prior technological connections between firms could influence which inventive regions an incursor entered, independently of the focal invention's effects. We therefore included three directed-pair controls: the number of citations from origin i to prospective destination j during the preceding fifteen years, the number of citations in the reverse direction, and the baseline directed search relationship from i toward j, measured as $\log\left[1/\left(1 + C_{ij}^-\right)\right]$. Both citation counts entered as $\log\left(1 + \text{count}\right)$. The firm-level controls in this specification described the prospective destination, since the model compared destinations for a fixed origin and focal invention.

4.4.4 Model Specification and Econometric Issues

Our three outcomes required different comparisons. To examine which firms undertook incursions and which became their destinations, we compared eligible firms associated with the same focal invention. This allowed each invention to have its own baseline level of incursion activity while relating differences among firms to their invention-induced changes in relative recombinant advantage. To examine destination selection, we instead compared eligible destinations for the same observed incursor around the same focal invention. This held constant the originating firm and the common invention while examining variation across prospective destinations.

Because each outcome was binary, we estimated three logistic regression models with group-specific fixed intercepts:

$$
\begin{aligned}
\Pr\left(Y_{ig}^O = 1\right) &= \Lambda\left(\alpha_g^O + \beta_O z\left(\Delta R_i^g\right) + \boldsymbol{\gamma}_O^\top \mathbf{X}_{ig}\right), \\
\Pr\left(Y_{jg}^T = 1\right) &= \Lambda\left(\alpha_g^T + \beta_T z\left(\Delta R_j^g\right) + \boldsymbol{\gamma}_T^\top \mathbf{X}_{jg}\right), \\
\Pr\left(Y_{ijg}^D = 1 \mid i \in I_g\right) &= \Lambda\left(\alpha_{ig} + \beta_D z\left(\Delta S_{ij}^g\right) + \boldsymbol{\gamma}_D^\top \mathbf{X}_{jg} + \boldsymbol{\delta}^\top \mathbf{Z}_{ijg}\right).
\end{aligned}
$$

Here, Λ denotes the logistic function, $\mathbf{X}$ contains the sixteen firm-level controls and applicable

missingness indicators, and **Z** contains the three directed-pair controls. The set I_g contains firms observed undertaking at least one qualifying incursion associated with invention g. The first two specifications include separate fixed intercepts for each focal invention, α_g^O and α_g^T, so their coefficients are estimated from differences among firms associated with the same invention. The third includes an origin-by-invention fixed intercept, α_{ig}, so its coefficient is estimated from differences among eligible destinations for the same observed incursor. It therefore examines where observed incursors enter, conditional on their having undertaken an incursion.

We standardized the substantive explanatory variables using fixed pooled-design means and standard deviations, denoted by z. The firm-level and directed-pair specifications used their respective retained reference samples, with scales fixed before removing fixed-effect groups without outcome variation. We applied these pooled scales rather than standardizing separately within inventions or origin–invention groups. Accordingly, the principal coefficients describe associations with a one-standard-deviation increase in the corresponding network-change measure, conditional on the included controls and fixed intercepts. Generated missingness indicators remained binary.

To limit the influence of extreme network changes, we retained firm observations within the pooled first-to-ninety-ninth percentile ranges of recombinant opportunity, exposure, and relative-advantage changes. We applied the corresponding restriction to destination-specific recombinant advantage changes in the directed-pair sample. These restrictions excluded observations outside the specified ranges without changing the retained values. We calculated opportunity, exposure, and the associated recombinant-advantage measures using the complete eligible network comparison populations before applying any observation-level restrictions.

We preserved observations with incomplete control information through pooled-median imputation and corresponding missingness indicators. For firms with a single historical utility patent, we assigned zero to the structurally undefined connectedness value in the estimation data and retained the separate one-patent indicator. Dependent-variable outcomes that remained unavailable were excluded rather than imputed. We also excluded fixed-effect groups containing only one observed outcome category, since such groups provided no within-group variation for estimating the hypothesized associations.

Finally, firms could participate in several focal-invention populations and appear repeatedly as origins or destinations. We therefore calculated dependence-adjusted standard errors accounting for shared focal inventions and repeated firms in the first two specifications. For destination selection, we also accounted for dependence arising when observations shared either firm, including firms appearing in different roles, with overlapping dependence contributions adjusted.

The resulting estimation samples contained 20,547 firm–invention observations for incursion origins, 20,756 for incursion destinations, and 28,375 directed firm-pair observations for destination selection. We report two-sided tests and 90% confidence intervals. Table 9 reports the three specifications together with their historical controls, allowing the origin, target, and destination-selection estimates to be compared within a consistent presentation.

5. Results

5.1 Relative Recombinant Advantage and Search Incursions

Our empirical analysis was designed as a deliberately conservative pilot test of the directional predictions of our theory. We restricted the representation of technological knowledge to an undirected patent-citation network and represented each firm by a single, fixed prior invention rather than its complete inventive portfolio. These choices excluded substantial sources of variation and minimized the discretionary assumptions entering our estimates of interfirm search costs. Our objective was to examine whether even this restricted representation of technological knowledge could recover the directional relationships predicted by our theory, rather than to estimate the full economic magnitude of invention-induced changes in competitive advantage.

Table 8 reports the changes in firms' recombinant opportunity, exposure, and relative recombinant advantage following the introduction of focal inventions. On average, recombinant opportunity increased by 1.065 percent, while exposure increased by 1.061 percent. Although the two changes were highly correlated (Figure 12), their relative magnitudes differed across firms. Some firms experienced greater proportional improvements in their ability to search toward other firms' established inventive regions, while others became disproportionately more exposed to incursions into their own. We examined whether these differences predicted the incidence and direction of subsequent search incursions.

Table 9 presents the results of three logistic regression models. Models 1 and 2 compare firms associated with the same focal invention, while Model 3 compares prospective incursive destinations for the same firm undertaking an incursion. All models include the historical controls and fixed effects described in the preceding section. We report odds ratios corresponding to one-standard-deviation changes in the explanatory measures and adjusted probability differences between their 25th and 75th percentiles within the relevant comparison groups.

Hypothesis 1 predicted that firms experiencing greater improvements in relative recombinant advantage would be more likely to undertake search incursions. Model 1 supports this prediction. The coefficient on relative recombinant advantage was positive and statistically significant at the 10 percent level. A one-standard-deviation improvement was associated with 8.8 percent higher odds of undertaking a search incursion (odds ratio = 1.088). The corresponding adjusted probability increased from 10.360 percent to 10.599 percent between firms at the 25th and 75th percentiles of relative recombinant advantage within the same focal invention, an increase of 0.239 percentage points. Figure 13 illustrates this relationship. Firms whose outward recombinant opportunities improved proportionately more than their exposure to incursions were thus more likely to extend their inventive activity into regions previously associated with other firms.

Hypothesis 2 concerned firms' susceptibility to incursions by others. We predicted that improvements in relative recombinant advantage would reduce the likelihood of becoming an incursion target. The estimates in Model 2 followed the predicted direction, although the statistical evidence was weaker than that obtained for Hypothesis 1. A one-standard-deviation improvement in relative recombinant advantage was associated with 4.6 percent lower odds of becoming an incursion target (odds ratio = 0.954). The adjusted probability similarly declined from 12.047 percent to 11.839 percent between

the 25th and 75th percentiles, a reduction of 0.208 percentage points. Both numerical estimates reflected the predicted negative association between relative recombinant advantage and susceptibility to incursions. The coefficient did not reach statistical significance at the 10 percent level, however, rendering the evidence for Hypothesis 2 suggestive rather than conclusive.

Hypothesis 3 examined the particular destinations of incursive search. An improvement in Firm A's overall relative recombinant advantage, accompanied by a deterioration in Firm B's, need not make Firm A particularly well positioned to undertake an incursion into Firm B's established inventive region. The attractiveness of this specific incursive relationship also depends on how the invention changes Firm A's cost of searching toward Firm B's established knowledge position. Model 3 provides support for this prediction. The coefficient on destination-specific relative recombinant advantage was positive and statistically significant at the 10 percent level. Among the eligible destinations of the same incursing firm, a one-standard-deviation improvement in destination-specific relative recombinant advantage was associated with 7.2 percent higher odds of an incursion into the corresponding firm's inventive region (odds ratio = 1.072). The adjusted probability increased from 9.478 percent to 9.518 percent between the 25th and 75th percentiles, an increase of 0.040 percentage points. Figure 14 illustrates this relationship. The findings suggest that invention-induced changes in relative recombinant advantage help explain both the propensity of firms to undertake search incursions and the particular firms toward which their incursive search is directed.

Several historical firm characteristics also exhibited systematic associations with incursion activity. Greater semantic distance between a firm's prior inventions and the focal invention was positively associated with undertaking search incursions, but negatively associated with becoming an incursion target or being selected as one. Larger prior patent stocks and greater technological breadth were negatively associated with becoming a target and with selection as an incursive destination. Baseline network scale exhibited opposing associations across the two roles, with firms possessing greater initial network scale being less likely to undertake incursions but more likely to become their targets. These findings indicate that firms' established technological positions continued to shape incursive activity alongside the relative recombinant advantages conferred by new inventions.

Table 11: Hypothesis Tests

Test	What we wanted to investigate	What we found and what it suggests
H1 · Undertaking incursions	Whether firms experiencing greater improvements in relative recombinant advantage become more likely to undertake search incursions.	+8.8% odds per SD improvement ($p < 0.10$). Firms whose outward inventive opportunities improved relative to their exposure to incursions were more likely to enter other firms' established inventive regions.
H2 · Becoming a target	Whether improvements in relative recombinant advantage reduce a firm's susceptibility to incursions by others.	–4.6% odds per SD improvement. Firms gaining relative advantage tended to become less susceptible to incursions. The relationship followed the predicted direction, with the 90% confidence interval narrowly including the null.
H3 · Selecting destinations	Whether destination-specific relative recombinant advantage predicts which particular firms become the targets of observed incursors.	+7.2% odds per SD improvement ($p < 0.10$). Invention-induced changes in the advantages of particular firm-to-firm relationships helped predict where incursors directed their inventive activity.

Table 12: Underlying Mechanisms

Test	What we wanted to investigate	What we found and what it suggests
Undertaking incursions	Whether incursive search responds to improvements in the opportunity-to-exposure balance or to general improvements in both dimensions of search.	Relative balance: +8.4% odds (p = 0.088). Common improvement: −6.7% (p = 0.294). The relative improvement in outward opportunity, rather than the common improvement in opportunity and exposure, exhibited the predicted positive association with undertaking incursions.
Becoming a target	Whether incursion susceptibility is associated with general improvements in recombinant opportunity and exposure or with changes in their relative balance.	Common improvement: +7.8% odds (p = 0.020). Relative balance: −3.4% (p = 0.225). Firms receiving incursions were more strongly associated with broad improvements in their search relationships, even when outward opportunities and incoming exposure improved at comparable rates.
Selecting destinations	Whether target selection is associated with changes in directed search costs, the prospective target's relative recombinant advantage, or both.	Directed search improvement: +3.0% odds (p = 0.220). Target advantage improvement: −5.2% odds (p = 0.067). Joint test: p = 0.013. The prospective target's changing relative position mattered alongside the direct search relationship between the firms. Both components were jointly associated with target selection.

Table 13: Predictive Importance (XGBoost)

Test	What we wanted to investigate	What we found and what it suggests
Undertaking incursions	How prominently our invention-induced measures feature alongside established predictors of which firms undertake incursions.	1.87% of total split gain. Historical knowledge structure (8.20%) and technological positioning (7.69%) received larger shares. Firms' established knowledge characteristics featured more prominently than invention-induced changes in the fitted model for incursion origins.
Becoming a target	How prominently invention-induced measures feature in predicting which firms' established inventive regions receive incursions.	14.72% of total split gain, the third-largest predictor group. Invention-induced changes in recombinant opportunity, exposure, and relative advantage featured prominently alongside firms' historical knowledge characteristics in distinguishing incursion targets.
Selecting destinations	How prominently invention-induced firm-level and directed search measures feature in predicting particular incursor–target relationships.	30.07% of total split gain, the largest displayed predictor group. Despite the deliberately restricted construction of our technological knowledge network, invention-induced measures featured prominently in the fitted model distinguishing particular incursive destinations, consistent with the relational emphasis of our theory.

Odds changes correspond to a one-standard-deviation improvement in the relevant measure. XGBoost percentages represent grouped split-gain shares in fitted models, rather than shares of outcomes explained or independent out-of-sample predictive contributions. The destination-selection model includes eight invention-induced predictors, compared with three in each firm-level model.

5.2 Distinguishing Changes in Recombinant Opportunity, Exposure, and Directed Search Costs

The results in Section 5.1 suggest that invention-induced changes in relative recombinant advantage help explain the incidence and direction of search incursions. We examined the components of these changes to understand more precisely the economic relationships underlying our findings. An invention may reduce the costs a firm faces in searching toward other firms' established inventive regions while simultaneously reducing the costs other firms face in searching toward its own. We sought to distinguish whether incursion activity was associated with improvements in both dimensions of search or, more specifically, with the extent to which outward recombinant opportunity improved relative to incoming exposure.

We separated these changes into two components. The first measured the average logarithmic improvement in recombinant opportunity and exposure, while the second captured the change in their relative balance:

$$M_{ig} = \frac{1}{2}\left[\log\left(\frac{O_i^g}{O_i^-}\right) + \log\left(\frac{E_i^g}{E_i^-}\right)\right]$$

$$D_{ig} = \log\left(\frac{O_i^g}{O_i^-}\right) - \log\left(\frac{E_i^g}{E_i^-}\right)$$

The distinction allowed us to examine whether the general improvement in a firm's outward opportunities and incoming exposure had consequences different from changes in its relative recombinant advantage. We entered both components simultaneously into the original incursion-origin and incursion-target regressions, retaining the historical controls, fixed effects, and dependence-adjusted standard errors. For destination selection, we further distinguished changes in the cost of searching toward a particular firm from changes in that prospective target's own opportunity-to-exposure balance. The invention-induced improvement in directed search from firm i toward firm j was measured as:

$$L_{ijg} = \log\left(\frac{1 + C_{ij}^-}{1 + C_{ij}^g}\right)$$

We entered this measure alongside the prospective target's change in relative recombinant advantage. Because the model compared potential destinations for the same incursor and focal invention, the incursor's own firm-level change was absorbed by the fixed effect. The additional models employed separately standardized logarithmic changes. Table 10 reports the estimates, and Figure 15 presents the corresponding adjusted probability contrasts.

Undertaking search incursions. The results suggest that the propensity to undertake an incursion is associated more closely with improvements in a firm's opportunity-to-exposure balance than with

the general improvement in both dimensions of search. The coefficient on the relative balance was positive and statistically significant at the 10 percent level ($\beta = 0.081,\ p = 0.088$). A one-standard-deviation improvement was associated with 8.4 percent higher odds of undertaking an incursion, corresponding to an adjusted probability increase of 0.231 percentage points between the 25th and 75th percentiles. By comparison, the average improvement in recombinant opportunity and exposure exhibited a negative but statistically insignificant association ($\beta = -0.069,\ p = 0.294$). These findings are consistent with our argument that the likelihood of undertaking an incursion depends on how much an invention improves a firm's outward search opportunities relative to the exposure confronting its established inventive position.

Becoming an incursion target. The results revealed a different pattern for firms receiving incursions. The average improvement in recombinant opportunity and exposure was positively associated with becoming a target ($\beta = 0.075,\ p = 0.020$). A one-standard-deviation improvement corresponded to 7.8 percent higher odds of receiving an incursion, while the adjusted probability increased by 0.351 percentage points between the 25th and 75th percentiles. The change in the opportunity-to-exposure balance retained the predicted negative association, although the estimate was less precise ($\beta = -0.035,\ p = 0.225$). This pattern suggests that an invention can increase a firm's susceptibility to incursions by reducing the costs other firms face in searching toward its established inventive region, even when the firm's own outward opportunities improve at a comparable rate. Consequently, a firm may become more susceptible to incursions without experiencing a substantial deterioration in its overall relative recombinant advantage. This distinction also helps explain the weaker statistical evidence for Hypothesis 2 in the combined specification.

Selecting incursive destinations. The destination-specific regressions examined whether particular firms were selected as targets because the invention reduced the incursor's costs of searching toward their established inventive regions, or because it altered the prospective targets' own relative recombinant advantages. The coefficient on the improvement in directed search was positive but statistically insignificant ($\beta = 0.030,\ p = 0.220$). By comparison, an improvement in the prospective target's opportunity-to-exposure balance was negatively associated with being selected as a destination ($\beta = -0.053,\ p = 0.067$). A one-standard-deviation improvement in the target's relative balance corresponded to 5.2 percent lower odds of receiving an incursion from the observed incursor, with an adjusted probability reduction of 0.134 percentage points. The two components were jointly associated with target selection (p = 0.013).

These destination-specific findings suggest that the attractiveness of an established inventive region depends on both the change in the incursor's cost of searching toward that region and the change in the prospective target's own relative recombinant advantage. The latter exhibited the more precisely estimated independent association in our regressions. Thus, even after accounting for changes in the costs of searching between particular firms, the invention-induced change in a prospective target's position relative to the wider ecology of inventing firms appears consequential for the direction of incursive search.

5.3 Predictive Importance of Recombinant Advantage and Competing Explanations

Our preceding analyses examined whether invention-induced changes in relative recombinant advantage were associated with search incursions and which components of these changes contributed to the observed relationships. We next examined the importance of our theoretical measures alongside established predictors of inventive search. This comparison is particularly relevant given the deliberately restrictive construction of our measures. By treating patent-citation relationships as undirected and representing each firm's prior inventive position through a single fixed invention, we excluded substantial information about the structure and direction of technological knowledge relationships. We therefore sought to determine whether measures derived from this parsimonious representation of technological knowledge would nevertheless feature prominently in predicting incursive activity alongside firms' historical knowledge characteristics and search practices.

We employed extreme gradient boosting (XGBoost), a tree-based machine-learning method that accommodates nonlinear relationships and interactions among predictors without requiring these relationships to be specified in advance (Chen and Guestrin, 2016). Such predictive approaches complement conventional regression analyses by allowing researchers to examine the information contributed by different sets of predictors under more flexible functional forms (Shmueli, 2010; Mullainathan and Spiess, 2017). We estimated separate classification models for undertaking search incursions, becoming an incursion target, and selecting particular incursive destinations.

For the first two outcomes, we included invention-induced changes in recombinant opportunity, exposure, and relative recombinant advantage. For destination selection, we included these three measures for both the originating and prospective target firms, together with changes in directed search costs and destination-specific recombinant advantage. We considered these measures alongside sixteen predictors representing established explanations of inventive search, including firms' historical knowledge diversity and structure, technological positioning, search depth and scope, boundary spanning, and entry into unfamiliar technologies (Stuart and Podolny, 1996; Rosenkopf and Nerkar, 2001; Katila and Ahuja, 2002). Additional controls captured firms' prior patent stocks, observed patenting ages, invention years, and technological sectors. We fitted five models for each outcome using folds separated by focal invention, with model depth selected using the training data. These predictive models employed a smaller set of controls than the explanatory regressions and did not include focal-invention fixed effects.

We assessed the importance of each predictor group through its share of total split gain, which measures the contribution of variables to improvements in the model's fitting objective. We aggregated these contributions across theoretically related predictors and averaged their shares across the five fitted models. Figure 16 presents the resulting comparisons. The shares describe the relative importance assigned to predictor groups during model fitting, allowing us to examine how prominently our invention-induced measures featured alongside established predictors of inventive search.

Undertaking search incursions. The recombinant-advantage measures accounted for 1.87 percent of total split gain in the model predicting which firms undertook incursions. Several established predictors received larger shares, including knowledge structure (8.20 percent), technological positioning (7.69 percent), search depth and scope (4.02 percent), and knowledge diversity (3.58 percent). Although the regression analyses identified a positive association between improvements in relative

recombinant advantage and undertaking incursions, the historical knowledge characteristics and search practices of firms featured more prominently in the fitted predictive model for this outcome.

Becoming an incursion target. The invention-induced measures assumed substantially greater prominence in predicting which firms received incursions, accounting for 14.72 percent of total split gain. This was the third-largest share among the predictor groups, following knowledge diversity (20.87 percent) and knowledge structure (15.96 percent), and exceeding technological positioning (13.33 percent). These findings complement the component analysis in Section 5.2, which indicated that invention-induced changes in recombinant opportunity and exposure were informative about firms' susceptibility to incursions. The measures describing how an invention altered firms' search conditions thus featured prominently alongside established knowledge characteristics in distinguishing firms whose inventive regions received incursions.

Selecting incursive destinations. The most striking result emerged in the model predicting which particular firms became the targets of observed incursors. The invention-induced measures accounted for 30.07 percent of total split gain, exceeding every competing predictor group displayed in Figure 16. Knowledge diversity accounted for 17.60 percent, knowledge structure for 10.99 percent, and technological positioning for 6.47 percent. The prominence of our measures in this model is particularly noteworthy given the deliberately restricted information used to construct them. Even with undirected technological relationships and a single fixed prior invention representing each firm, the estimated changes in firms' recombinant positions and directed search costs accounted for the largest share of split gain among the displayed groups. The result is consistent with the central relational proposition of our theory, that predicting where a firm directs its incursive search requires information about how a particular invention changes the technological relationships between that firm and its prospective targets.

The recombinant-advantage group contained eight predictors in the destination-selection model, compared with three in each firm-level model, and the reported shares reflect their collective use in fitting the models rather than their independent contributions to out-of-sample predictive performance. Within this scope, the results reveal an informative pattern. Our invention-induced measures received relatively limited importance in predicting which firms undertook incursions, greater importance in predicting which firms became targets, and their greatest prominence in distinguishing the particular destinations of incursive search.

6. Discussion and Conclusions

An invention can alter the competitive significance of firms' established technological knowledge even before those firms have undertaken any new inventive activity themselves. By introducing new relationships among existing knowledge components, an invention can change the costs of searching between firms' established inventive positions, creating opportunities for some firms to develop inventions in regions previously associated with others. We developed a structural theory of these changes and examined how the resulting redistribution of recombinant advantage shaped firms' propensity to undertake search incursions, their susceptibility to incursions by others, and the particular destinations of incursive search.

Our empirical analysis revealed several patterns consistent with this account. Firms experiencing greater improvements in relative recombinant advantage were more likely to undertake search incursions, while observed incursors were more likely to enter inventive regions toward which their destination-specific relative recombinant advantage improved. The estimates concerning susceptibility to incursions also followed the predicted direction. The additional analyses revealed an important distinction in the changes underlying these outcomes. Undertaking incursions was associated with improvements in outward recombinant opportunity relative to incoming exposure, whereas becoming an incursion target was more strongly associated with the general improvement in recombinant opportunity and exposure. These findings bear on three related questions concerning the changing structure of local search, the different roles firms occupy in incursive competition, and the direction of inventive activity toward particular other firms.

6.1 Inventions and the Changing Structure of Local Search

Research on recombinant innovation has established that opportunities for invention depend on the availability of knowledge components and the technological relationships among them (Kogut and Zander, 1992; Fleming, 2001; Fleming and Sorenson, 2001, 2004). New inventions can make previously infeasible combinations possible or alter the economic attractiveness of combinations already within firms' technological reach. We extended this perspective by examining how the opportunities arising from such developments are distributed across firms occupying different positions within the broader structure of technological knowledge.

An important implication of our theory is that an invention can change what constitutes local search for different firms. By connecting knowledge components that were previously separated, an invention can alter the effort required to reach particular bodies of technological knowledge from particular firms' established inventive positions. These changes need not be uniform across firms or equivalent in opposite directions. An invention may substantially reduce the cost of searching from Firm A's prior knowledge position toward Firm B's established inventive region while producing a smaller change in the reverse direction or in Firm A's costs of searching toward other firms. The same technological development can therefore redistribute the opportunities for inventive search among firms whose accumulated knowledge remains unchanged.

The empirical findings illustrate the importance of this two-sided transformation. Following the introduction of focal inventions, recombinant opportunity and exposure increased by 1.065 and 1.061 percent on average, respectively. Although these changes were highly correlated, their relative magnitudes differed across firms. An invention that improved a firm's ability to reach other firms' established inventive regions could simultaneously reduce the costs other firms faced in searching toward its own. Consequently, an improvement in outward recombinant opportunity did not necessarily confer a comparable improvement in the firm's relative recombinant advantage.

These findings extend research on technological positioning, which has examined how firms' historical inventive activities locate them within technological space and condition their subsequent inventive development (Jaffe, 1986; Stuart and Podolny, 1996; Aharonson and Schilling, 2016). We examined how inventions originating elsewhere in the technological knowledge network altered the competitive significance of firms' established inventive positions. The historical knowledge positions of two

firms could remain unchanged while an external invention altered the costs of searching between them. The advantages associated with those positions therefore depended not only on the knowledge firms had accumulated, but also on the evolving relationships among the knowledge underlying inventions across the broader technological landscape.

The distinction also bears on research concerning technological discontinuities and changes in established competitive positions (Tushman and Anderson, 1986; Henderson and Clark, 1990; Tripsas, 1997). An invention need not constitute an industry-defining technological discontinuity to disturb the distribution of inventive advantages among firms. Particular connections introduced into the technological knowledge network can change the feasibility or economic attractiveness of entering established inventive regions, even while the broader technological architecture remains recognizable. Our framework provides a basis for examining how the continuous arrival of consequential inventions alters the conditions of inventive competition among particular firms.

6.2 Recombinant Opportunity, Exposure, and Firms' Roles in Search Incursions

The invention-induced redistribution of search costs has different implications for firms undertaking incursions and firms whose established inventive regions become their destinations. For a prospective incursor, the relevant consideration is the improvement in its opportunities to pursue inventive activity beyond its established region. For a prospective target, the relevant consideration is the reduction in costs other firms face in undertaking inventive activity within its established region. The distinction is consequential because the same invention can improve both dimensions of search without improving them by equivalent proportions.

For firms undertaking incursions, we found that greater improvements in relative recombinant advantage were associated with a higher likelihood of incursive search. The additional regressions sharpened this finding. When we separated the common improvement in recombinant opportunity and exposure from the change in their relative balance, the latter retained the predicted positive association with undertaking incursions. The common improvement in both dimensions did not exhibit a corresponding positive association.

This pattern is consistent with the resource-allocation reasoning underlying our theory. A new invention may reduce the costs of reaching inventive regions previously associated with other firms, expanding the opportunities available for outward search. Yet the same technological development can also increase other firms' opportunities to enter the focal firm's established inventive region. Since extending inventive activity and preserving established inventive positions both place demands on inventive resources, the relative improvement in outward opportunity can be consequential for the attractiveness of undertaking an incursion. Our findings suggest that the distribution of these changes, rather than the general expansion of recombinant opportunities alone, is relevant to understanding which firms direct inventive effort toward other firms' established regions.

The results concerning incursion targets revealed a different pattern. Improvements in relative recombinant advantage were associated with a lower estimated likelihood of becoming a target, in the predicted direction. The component regressions, however, identified a stronger positive association between the common improvement in recombinant opportunity and exposure and the likelihood of receiving an incursion. The change in the opportunity-to-exposure balance retained its predicted

negative association.

An invention can therefore increase a firm's susceptibility to incursions even when its own outward opportunities improve alongside its incoming exposure. Consider a firm whose established inventive region becomes easier for several other firms to reach following an external invention. The same invention may also reduce the costs the focal firm faces in searching toward other firms' inventive regions, leaving its opportunity-to-exposure balance largely unchanged. Nevertheless, prospective incursors now face lower costs of undertaking inventive activity within the firm's established region. The increase in incoming exposure can consequently be associated with a greater likelihood of receiving an incursion without requiring a substantial deterioration in the firm's overall relative recombinant advantage.

The distinction suggests that undertaking and receiving incursions are associated with different aspects of the same technological disturbance. Originating an incursion involves directing a firm's inventive activity toward another firm's established region, an outcome associated in our analysis with improvements in the originating firm's relative opportunities. Becoming a target reflects inventive activity undertaken by other firms and may be more closely associated with broader reductions in the costs of searching toward the target's established knowledge position. These roles need not characterize permanent classes of firms. The same firm can acquire new opportunities for outward invention while simultaneously becoming more exposed to incursions, depending on the invention under consideration and its relationships with the knowledge positions occupied by other firms.

6.3 Destination-Specific Recombinant Advantage and the Direction of Incursive Search

Identifying the firms that gain opportunities to undertake incursions and those that become more susceptible leaves open the question of which particular firms are likely to enter which others' established inventive regions. Research on innovation search has developed substantial explanations of how firms depart from familiar technological knowledge, cross organizational and technological boundaries, and vary the depth and scope of inventive search (Ahuja and Lampert, 2001; Rosenkopf and Nerkar, 2001; Katila and Ahuja, 2002). We extended this understanding by examining the destinations of inventive activity relative to the prior inventive positions of other firms.

An improvement in Firm A's overall relative recombinant advantage, accompanied by a deterioration in Firm B's, need not make Firm A particularly well positioned to undertake an incursion into Firm B's established inventive region. Firm A's improved opportunities may be concentrated toward Firm C, while Firm B's increased exposure may arise primarily from Firm D. Whether an incursion from Firm A into Firm B becomes more attractive therefore depends on three changes induced by the focal invention: the change in Firm A's overall opportunity-to-exposure balance, the corresponding change in Firm B's balance, and the change in the expected cost of searching from Firm A's knowledge position toward Firm B's established inventive region.

We incorporated these changes into a destination-specific measure of relative recombinant advantage, permitting an invention-by-invention examination of particular relationships between prospective incursors and targets. The empirical results supported the predicted relationship. Among firms already observed undertaking incursions, prospective targets toward which the originating firm's destination-specific relative recombinant advantage improved were more likely to receive incursions.

These estimates compared eligible destinations for the same incursing firm following the same focal invention, allowing us to examine the particular directions in which observed incursors extended their inventive activity.

The component regressions provided further insight into this relationship. Invention-induced reductions in the expected cost of searching from the incursing firm's knowledge position toward a prospective target's position exhibited the predicted positive association with target selection. Improvements in the prospective target's own opportunity-to-exposure balance, meanwhile, were negatively associated with receiving an incursion from the observed originator. The two components were jointly associated with target selection, with the prospective target's relative recombinant advantage exhibiting the stronger individual association.

The findings suggest that the attractiveness of an inventive destination depends on the technological relationship connecting the incursor to the prospective target, as well as the opportunities and exposure confronting that target in the wider technological knowledge network. An invention may substantially improve an incursor's ability to reach a particular inventive region while simultaneously conferring favorable changes in relative recombinant advantage upon the firm already active there. Another prospective target may experience a different combination of changes in directed search costs and relative advantage, making its established inventive region a more attractive destination for incursive search.

The direction of inventive competition therefore cannot be inferred solely from the overall advantages conferred upon individual firms. We extended the analysis of inventive search from the conditions favoring entry into other firms' established inventive regions to the particular firm-to-firm relationships in which such entry occurred. This provides a basis for explaining why an invention favors an incursion by one particular firm into another's established inventive region, even when other firms experience comparable changes in their overall recombinant opportunities or exposure.

6.4 Search Incursions as Competitive Microinteractions

The ability to identify particular firm-to-firm incursions also provides a basis for examining how individual inventions contribute to broader transformations in technological competition. Research on technological convergence, technological discontinuities, and changes in established competitive positions has developed substantial accounts of how technological developments reshape the fortunes of firms and industries (Rosenberg, 1963; Tushman and Anderson, 1986; Rothaermel and Hill, 2005; Uzunca, 2018). Search incursions offer a finer-grained unit of analysis for understanding the competitive microinteractions through which such transformations may emerge.

An invention can alter the costs of searching between numerous pairs of firms, changing which established inventive regions become attractive destinations for subsequent inventive activity. Firms acting upon these opportunities may develop inventions in regions previously associated with other firms, thereby changing the distribution of inventive activity across the technological landscape. Subsequent inventions can introduce further connections among knowledge components, altering the opportunities and exposure confronting the firms involved. Through the accumulation of these microinteractions, established technological advantages may be extended, penetrated, or gradually eroded, contributing to broader patterns of technological convergence and competitive change.

The framework thus offers a basis for examining the evolution of technological competition at the level of particular inventions and particular relationships between firms. A broad technological transformation may eventually become visible through changes in industry structure, the displacement of established technologies, or the redistribution of firms' innovative capabilities. The underlying competitive changes, however, can begin with individual inventions that alter the conditions under which one firm can undertake inventive activity in another firm's established region. Understanding these changes while they are still unfolding may permit firms and researchers to recognize emerging competitive opportunities and vulnerabilities before their broader consequences have fully materialized.

The predictive analyses provided a complementary indication of the relevance of this relational perspective. The invention-induced measures accounted for 30.07 percent of total split gain in the destination-selection model, the largest share among the displayed predictor groups. Their prominence in the fitted model was noteworthy given our deliberately restricted representation of technological knowledge, which employed undirected citation relationships and a single fixed prior invention to represent each firm. The measures received smaller shares in the models predicting incursion origins and targets, suggesting that their greatest prominence in these fitted models concerned the particular directions of incursive search. The grouped shares describe the use of these measures in fitting the models, with the destination-selection group containing more predictors than the firm-level groups.

The relational account also has implications for the strategic decisions confronting firms during technological change. The significance of an external invention for a particular firm may depend on the search relationships the invention alters, including relationships with firms operating in technological areas that previously appeared distant from its established knowledge. A firm evaluating a new technological development can therefore consider which established inventive regions have become less costly for it to reach, which other firms face lower costs of searching toward its own inventive region, and how these changes affect its relative recombinant advantage. Such an assessment may inform the direction of future inventive search, the identification of prospective technological competitors, and decisions concerning the preservation or extension of established inventive advantages.

The strategic value of understanding these microinteractions extends beyond anticipating the immediate inventive response to a particular technological development. A search incursion may introduce new intellectual property claims, competing technological approaches, or additional inventive participants into an area where another firm has accumulated knowledge and capabilities (Stuart and Podolny, 1996; Ziedonis, 2004; Katila and Chen, 2008). Recognizing how external inventions alter the prospects for such incursions can therefore inform firm-specific strategic action while broader changes in technological competition remain in the course of formation.

6.5 Scope and Directions for Further Research

Our empirical analysis was designed as an initial test of the directional predictions of our theory under a deliberately restricted representation of technological knowledge. We constructed an undirected patent-citation network and represented each firm's established knowledge position through a single

fixed prior invention. We then used a model of purposeful but error-prone search to estimate the minimum expected costs of reaching particular knowledge positions before and after the introduction of focal inventions. These choices limited the information and discretionary assumptions entering the construction of the interfirm search-cost measures. Extending the framework to incorporate directed technological relationships, richer representations of firms' historical inventive portfolios, and additional sources of technological knowledge would allow future research to examine how these features affect the estimated distribution of recombinant advantage. Evidence concerning research expenditures, inventor mobility, and the organization of inventive activity could further illuminate how firms allocate resources in response to the changing opportunities and exposure identified by the model.

The identification of search incursions presents a second opportunity for extending the analysis. We employed patent text and synthetic-patent constructions to identify subsequent inventions that departed from their originating firms' established inventive concentrations and entered regions associated with other firms' prior inventive activity. The sensitivity analyses showed that the relationship between relative recombinant advantage and originating incursions became weaker under broader definitions of incursive entry, whereas the destination-specific relationship remained more consistent across the examined thresholds. These findings raise the possibility that the conditions associated with pronounced incursions differ from those governing more moderate movements into other firms' established inventive regions. Subsequent research could examine the intensity of incursive search and employ alternative representations of inventive content, including patent claims, technological functions, and expert assessments of the underlying inventions. Such analyses could also help distinguish the substantive effects of firms' historical knowledge characteristics from relationships arising through the incursion-classification procedure.

Our analysis also examined a deliberately defined population of inventions and firms. We identified eligible firms through subsequent citations to focal inventions and estimated firm-level incursion relationships within invention-specific populations containing observed incursions. The resulting estimates therefore concerned firms with documented subsequent technological relationships to particular inventions, rather than every firm that might have been affected by those inventions. A broader design could identify prospective firms using information available before the focal invention and investigate why some inventions generate search incursions while others do not. Such an extension would permit the analysis to examine both the emergence of incursive activity around particular inventions and the distribution of that activity across prospective incursors and targets.

Finally, we examined the occurrence and direction of inventive entry into other firms' established inventive regions. The subsequent competitive consequences of these incursions constitute a distinct question. An incursion may alter the appropriability of existing inventions, encourage further inventive activity, or change the competitive relevance of previously accumulated technological capabilities. Its consequences may depend on the value of the resulting inventions, the complementary assets possessed by the firms involved, and their subsequent strategic actions. Linking observed incursions to changes in inventive performance, technological diversification, intellectual property disputes, licensing arrangements, and the movement of inventors would help establish when incursive entry produces competitive displacement and when it contributes to the broader development of technological knowledge.

6.6 Conclusion

Each invention can change what constitutes local technological knowledge for different firms. By introducing new relationships among existing knowledge components, an invention can alter the costs of searching between established inventive positions, redistributing firms' opportunities to undertake search incursions and their exposure to incursions by others. These changes depend on the positions firms occupy relative to the invention and to one another, giving rise to different consequences for particular firms and particular prospective incursive destinations.

We developed a structural theory of these relationships and examined their implications for the incidence and direction of search incursions. Our findings connected invention-induced changes in relative recombinant advantage to firms' subsequent inventive activity, while revealing differences in the changes associated with undertaking incursions, becoming their targets, and selecting particular destinations. The resulting framework advances toward a firm-by-firm, invention-by-invention account of technological competition, in which the consequences of new inventions can be traced through the particular competitive microinteractions they enable. Understanding these interactions provides a basis for examining how technological change continuously reshapes the distribution of inventive opportunities, and how broader transformations in technological competition emerge from changes in the relationships among firms.

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Additional Tables and Figures

Additional Tables (Tables 4–10)

Table 4: Characteristics of the focal influential inventions

Characteristic	Full corporate sample	Network sample
A. Coverage		
Influential inventions	5,123	1,727
Distinct adopting firms	24,606	4,824
Firm-invention observations	90,930	21,646
Composite ranking score	0.63 (0.19)	0.61 (0.19)
Adopting firms per invention	17.75 (12.87)	12.53 (8.12)
Inventors per invention	2.55 (1.78)	2.55 (1.77)
Backward technological breadth	0.74 (0.12)	0.74 (0.12)
B. Grant cohort: number of focal inventions (%)		
1985-1989	786 (15.3%)	229 (13.3%)
1990-1994	1,103 (21.5%)	397 (23.0%)
1995-1999	1,271 (24.8%)	389 (22.5%)
2000-2004	1,441 (28.1%)	523 (30.3%)
2005-2009	522 (10.2%)	189 (10.9%)
C. Technology sector: number of focal inventions (%)		
Electrical engineering	2,343 (45.7%)	878 (50.8%)
Multiple sectors	1,337 (26.1%)	430 (24.9%)
Instruments	724 (14.1%)	229 (13.3%)
Mechanical engineering	372 (7.3%)	99 (5.7%)
Chemistry	288 (5.6%)	72 (4.2%)
Other fields	54 (1.1%)	18 (1.0%)
Unclassified	5 (0.1%)	1 (0.1%)

Note. Continuous entries are means (standard deviations); cohort and sector entries are counts (percentages). The full sample covers inventions granted in 1985-2009 and corporate citing firms observed during the next fifteen years. The network sample retains inventions with an observed incursion and firms with eligible prior representatives. Technology sectors describe patents, not firm industries.

Table 5: Illustrative selected and excluded inventions

Patent	Title	Grant year	Composite score
Affirmative classifications retained in the focal sample			
US 7,393,699	NANO-electronics	2008	1.226
US 6,850,820	Distributed power generation system, and maintenance system and maintenance method utilizing the same	2005	1.154
US 5,801,073	Net-shape ceramic processing for electronic devices and packages	1998	1.152
US 5,215,820	Metal-electroconductive polymer composite fine particles and method for preparing same	1993	1.131
US 5,216,674	Method of and device for bringing a network interface out of a sleep mode into a wake-up state	1993	1.127
US 5,617,515	Method and apparatus for controlling and programming a robot or other moveable object	1997	1.118
US 6,020,419	Transparent coating compositions containing nanoscale particles and having improved scratch resistance	2000	1.102
US 5,801,070	High sensitivity, silicon-based, Microcalorimetric gas sensor and fabrication method	1998	1.099
Negative classifications excluded from the focal sample			
US 5,800,897	Air freshener composition containing a fiber pad	1998	1.034
US 4,852,843	Beverage holder for attachment to vehicle heating and cooling vents	1989	0.958
US 6,457,883	Interactive printer reward scheme	2002	0.947
US 4,728,143	Door for vehicle which incorporates loudspeaker	1988	0.923
US 4,597,218	Sachet for use in pest control	1986	0.922
US 5,290,063	Hand grip construction for a ski pole or the like	1994	0.910
US 4,690,299	Bulk carbonated beverage container	1987	0.905
US 4,579,762	Stain resistant carpet with impervious backing	1986	0.894

Note. All sixteen examples fall among the first 6,000 reviewed candidates and were granted in 1985-2009. Entries reproduce the original classification and composite ranking score. They illustrate the screen; they are not a matched comparison or a new validation.

Table 6: Sample construction and observed search incursions

Sample or statistic	Count
A. Corporate adoption universe	
Focal inventions	5,123
Distinct adopting firms	24,606
Firm-invention observations	90,930
Adoption links (invention-firm-patent)	263,680
Scored adoption links	139,644
Unscored observations	124,036
Of which: no own-firm priors	122,068
Firm-invention observations with at least one score	50,658
B. Elbow-defined incursion activity	
Focal inventions with an observed incursion	1,727
Distinct Incursion patents	3,888
Incursion events	4,212
Distinct origin firms	1,346
Distinct attributed target firms	1,197
Distinct firms observed in both roles	244
Distinct invention-origin-target pairs	2,773
C. Network and regression coverage	
Full-network firm-invention observations	21,646
Full-network directed pairs	363,390
Retained firm-invention descriptive observations	21,102
Retained observations with known origin status	20,725
Retained known-origin directed pairs	349,353

Note. An observation links a focal invention, a citing firm, and a citing patent. The same patent may appear around more than one invention. Scores of at least 0.709 identify incursions. Each event is assigned one target. Unscored observations remain unavailable. Network measures are calculated before trimming; the retained samples exclude extreme network changes.

Table 7: Prior inventive characteristics by observed firm role

Characteristic	All firms	Origin only	Target only	Both	Neither
Firm-invention observations	21,102	2,123	2,402	36	16,164
Prior patent stock	3,063 (6,378)	853 (3,881)	2,519 (5,791)	1,184 (5,582)	3,455 (6,675)
Patenting age (years)	12.7 (8.7)	6.8 (7.5)	12.2 (8.2)	5.3 (7.6)	13.6 (8.5)
Patenting momentum	0.90 (1.14)	0.38 (0.65)	0.89 (1.10)	0.37 (0.59)	0.98 (1.18)
Prior-patent originality	0.41 (0.16)	0.42 (0.23)	0.40 (0.16)	0.42 (0.25)	0.41 (0.15)
Knowledge-base connectedness	0.10 (0.17)	0.11 (0.22)	0.09 (0.17)	0.05 (0.13)	0.09 (0.17)
Technological breadth	0.74 (0.30)	0.35 (0.37)	0.73 (0.28)	0.20 (0.34)	0.79 (0.25)
Semantic distance to focal invention	0.82 (0.13)	0.92 (0.11)	0.81 (0.14)	0.92 (0.09)	0.81 (0.13)
Prior search scope	0.74 (0.22)	0.86 (0.23)	0.74 (0.21)	0.89 (0.21)	0.73 (0.21)
Unavailable origin status	377	0	0	0	0

Note. Entries are means (standard deviations), using available values before imputation. Roles are mutually exclusive within an invention. The all-firms column includes 377 observations with unavailable origin status; they are not assigned to the four role groups. A target is an attributed destination, not evidence of economic harm. All histories precede the focal invention.

Table 8: Network positions and invention-induced changes

Measure	N	Mean	SD	Median
A. Firm-invention positions before insertion				
Opportunity: outward accessibility	21,102	2.734	2.388	2.168
Exposure: inward accessibility	21,102	2.734	2.393	2.169
Opportunity-to-exposure balance	21,102	2,210.766	37,791.730	1.000
B. Invention-induced changes				
Opportunity change (%)	21,102	1.065	2.637	0.011
Exposure change (%)	21,102	1.061	2.643	0.003
Relative recombinant advantage (bp)	21,102	0.467	7.438	0.022
Directed accessibility change (bp)	349,353	109.387	483.137	0.000
Destination-specific recombinant advantage (bp)	349,353	109.964	483.132	0.025

Note. Opportunity sums outward access and exposure sums inward access over the same eligible firms. Access equals one divided by one plus directed search cost. The balance is opportunity divided by exposure. Destination-specific advantage combines directed access with the origin-to-target ratio of balances. Changes are after/before minus one; one basis point (bp) equals 0.01 percent.

Table 9: Logistic models of incursion origin, target, and destination selection

Predictor	(1) Origin	(2) Target	(3) Destination selection
Network predictor			
Relative recombinant advantage	0.085* (0.046)	-0.047 (0.030)	--
Destination-specific recombinant advantage	--	--	0.070* (0.038)
Prior firm characteristics			
Prior patent stock	-0.196 (0.149)	-0.389*** (0.086)	-0.362*** (0.088)
Patenting age	-0.129 (0.100)	0.097 (0.075)	0.071 (0.075)
Patenting momentum	0.032 (0.089)	-0.032 (0.062)	-0.062 (0.060)
Prior-patent originality	0.076 (0.046)	-0.043 (0.041)	-0.034 (0.040)
Knowledge-base connectedness	-0.060 (0.050)	0.005 (0.044)	0.026 (0.043)
One-patent portfolio	0.214*** (0.053)	-0.154*** (0.058)	-0.054 (0.055)
Technological breadth	0.087 (0.092)	-0.212*** (0.071)	-0.193*** (0.070)
Semantic distance to focal invention	0.217*** (0.080)	-0.206*** (0.038)	-0.222*** (0.036)
Selected prior portfolio size	-1.421*** (0.145)	0.242*** (0.084)	0.115 (0.083)
Prior search depth	0.326*** (0.106)	-0.126 (0.085)	-0.096 (0.090)
Prior search scope	-0.383*** (0.103)	-0.046 (0.079)	-0.042 (0.083)
External-citation share	-0.229*** (0.085)	-0.015 (0.042)	-0.000 (0.038)
Knowledge-base vintage	0.118 (0.081)	-0.063 (0.054)	-0.068 (0.053)
External-source nonredundancy	-0.164*** (0.048)	0.081** (0.039)	0.071* (0.040)
Baseline network scale	-1.249*** (0.212)	0.526*** (0.157)	0.535*** (0.181)
Baseline opportunity-to-exposure balance	-0.163*** (0.049)	-0.012 (0.058)	-0.185 (0.138)
Pair controls and estimation sample			
Prior origin-to-target citations	--	--	-0.158 (0.122)
Prior target-to-origin citations	--	--	-0.140 (0.092)
Baseline directed accessibility	--	--	-0.179 (0.153)
Observations	20,547	20,756	28,375
Positive outcomes	2,152	2,483	2,701
Focal inventions	1,701	1,674	1,678
Distinct firms	4,699	4,727	4,758
Fixed effects	Invention	Invention	Origin by invention
Missingness indicators	Yes	Yes	Yes

Note. Entries are log-odds coefficients (standard errors). Substantive predictors use fixed pooled-design scales. Model 3 includes target characteristics and compares targets within origin-invention groups. Missing controls are median-imputed with indicators. Standard errors account for shared inventions and repeated firms; Model 3 also allows either endpoint to repeat. Constant-outcome groups are excluded. * p < .10; ** p < .05; *** p < .01, two-sided.

Table 10: Component models of opportunity, exposure, and directed access

Component or contrast	Origin	Target	Destination selection
A. Unrestricted log-change components			
Average access improvement	-0.069 (0.066)	0.075** (0.032)	--
Opportunity-to-exposure balance change	0.081* (0.048)	-0.035 (0.029)	--
Directed access improvement	--	--	0.030 (0.025)
Target's balance change	--	--	-0.053* (0.029)
B. Substantive contrasts: own-group 25th to 75th percentile			
Average access improvement (pp)	-0.203 [-0.517, +0.111]	+0.351 [+0.103, +0.600]	--
Balance change (pp)	+0.231 [+0.007, +0.455]	-0.161 [-0.376, +0.054]	--
Directed access improvement (pp)	--	--	+0.027 [-0.009, +0.064]
Target's balance change (pp)	--	--	-0.134 [-0.251, -0.017]
C. Diagnostics			
Joint component test: p-value	0.102	0.013	0.013
Average access improvement: p-value	0.294	0.020	--
Equal-and-opposite component coefficients: p-value	--	--	0.082
Observations	20,547	20,756	28,375
Original controls / fixed effects	Yes	Yes	Yes

Note. Panel A gives standardized log-odds coefficients (standard errors). Panel B gives within-group 25th-to-75th percentile probability differences [90% confidence intervals], in percentage points. The other component and controls remain at observed values. Controls, fixed effects, and dependence corrections match Table 9. The equal-and-opposite restriction concerns unstandardized log components. * p < .10; ** p < .05; *** p < .01, two-sided.

Additional Figures (Figures 6–16)

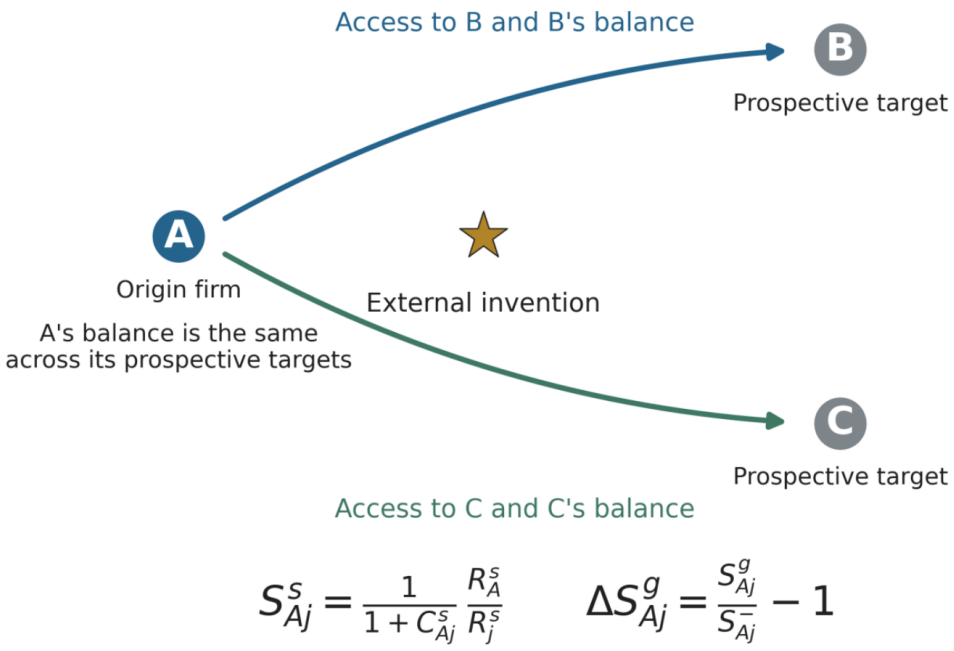


Figure 6: Destination-specific recombinant advantage

Note. For a given origin and invention, the origin's opportunity-to-exposure balance is common across targets. Destinations differ in directed search cost and their own balances. The index combines these three terms; its change compares the network with and without the invention. Arrows are schematic search relationships, not observed incursions.

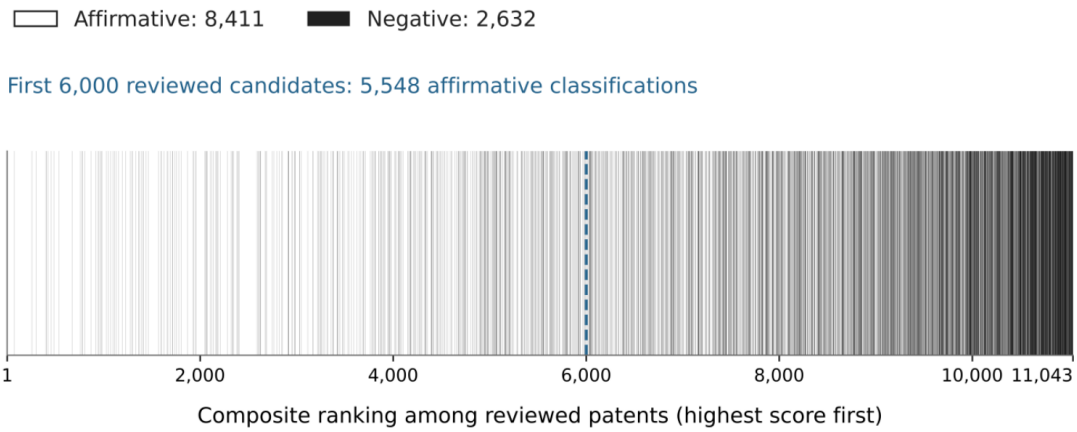


Figure 7: Influential-invention classifications across the composite ranking

Note. Each strip is one of 11,043 reviewed patents. White and black indicate affirmative and negative LLM classifications. The dashed line marks the first 6,000 reviewed candidates. Of their 5,548 affirmative classifications, 5,123 fall in the 1985-2009 study period. Classifications and ordering are unchanged from the original screen.

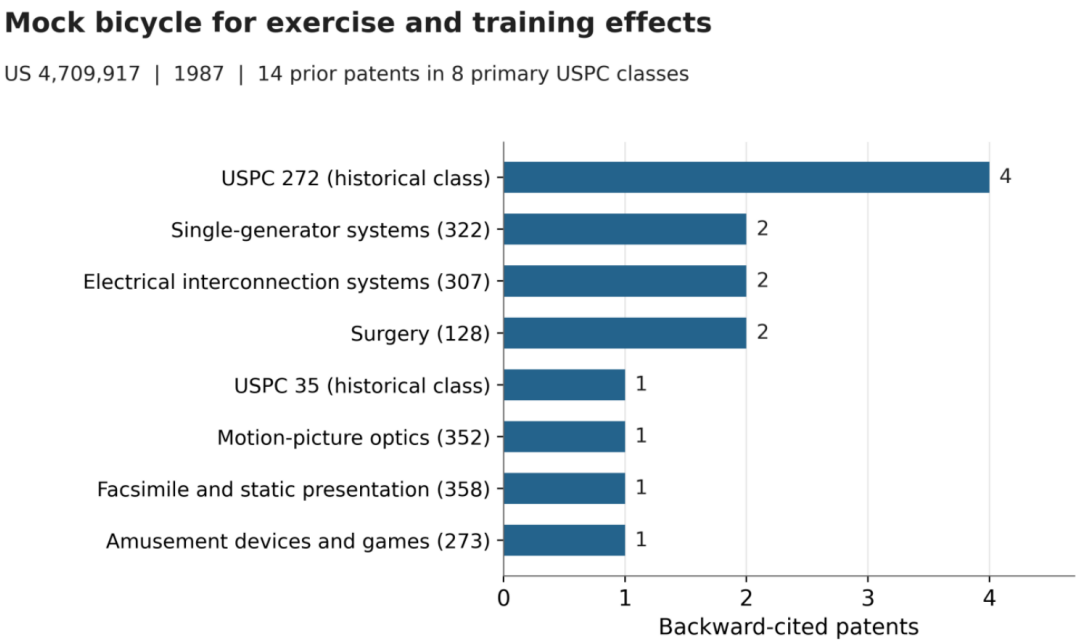


Figure 8: Technological diversity of an invention's backward citations

Note. The example shows the 14 backward-cited patents in the saved fifteen-year historical network for US 4,709,917. Bars count patents by primary three-digit USPC class. It is a window-restricted illustration, not the full 25-reference list used in the patent record. Retired class codes are retained rather than assigned new names.

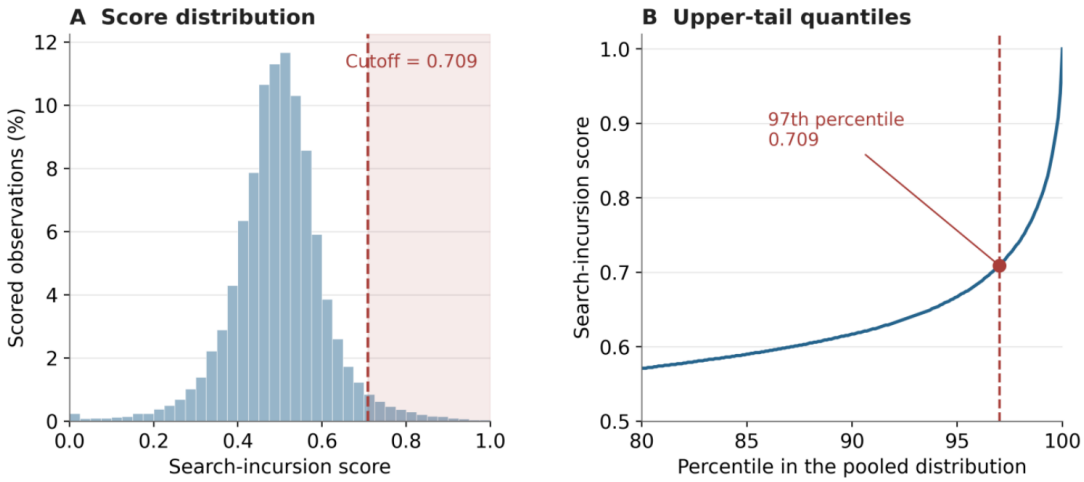


Figure 9: Search-incursion scores and the classification threshold

Note. The sample contains 139,644 scored invention-firm-patent observations. The score is the fraction of 1,000 simulations in which the external text pool more closely matches the citing patent than the own-firm pool. The 97th-percentile cutoff, 0.709, lies midway between the two detected elbow percentiles. Scores at or above the cutoff identify 4,212 events (3.02%).

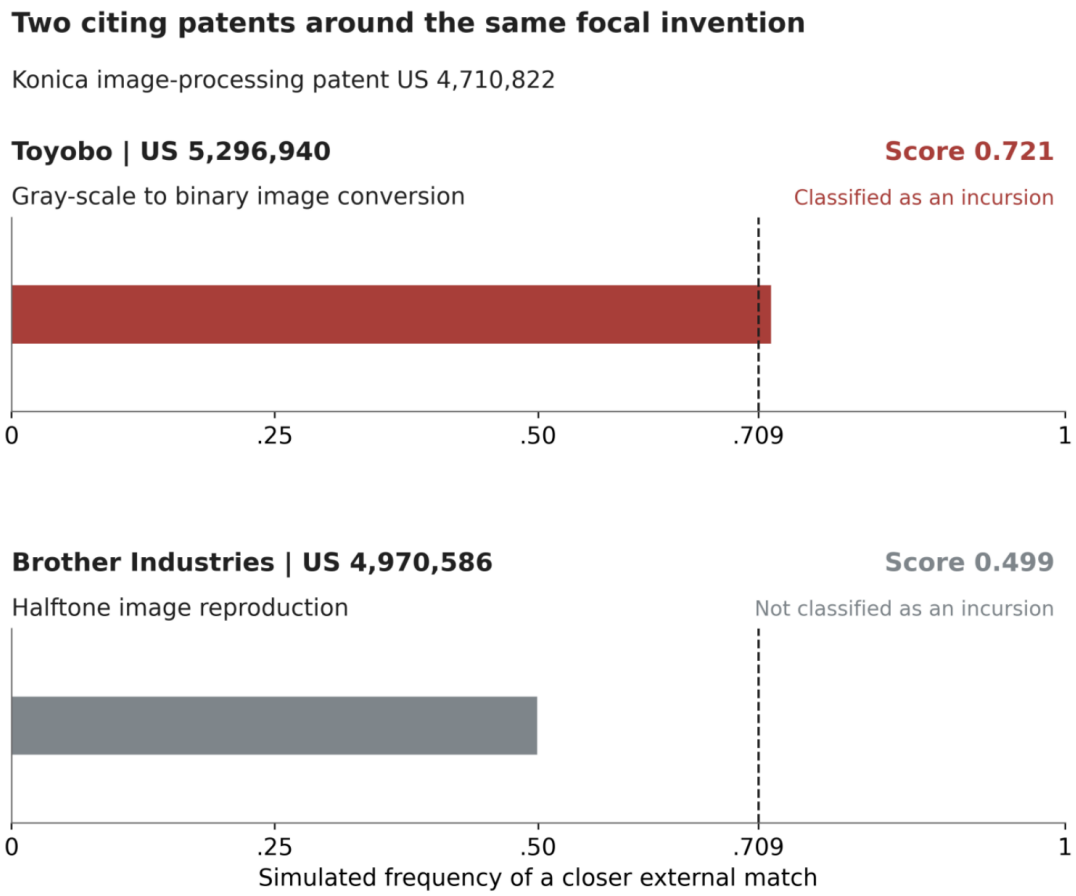


Figure 10: Examples of the search-incursion classification

Note. Both citing firms have twenty selected prior patents. The bars show their saved simulated comparison frequencies; the dashed line is the pooled cutoff of 0.709. Toyobo's patent is above the cutoff and Brother Industries' patent is below it. These illustrate the classification rule, not an independent validation of the two cases.

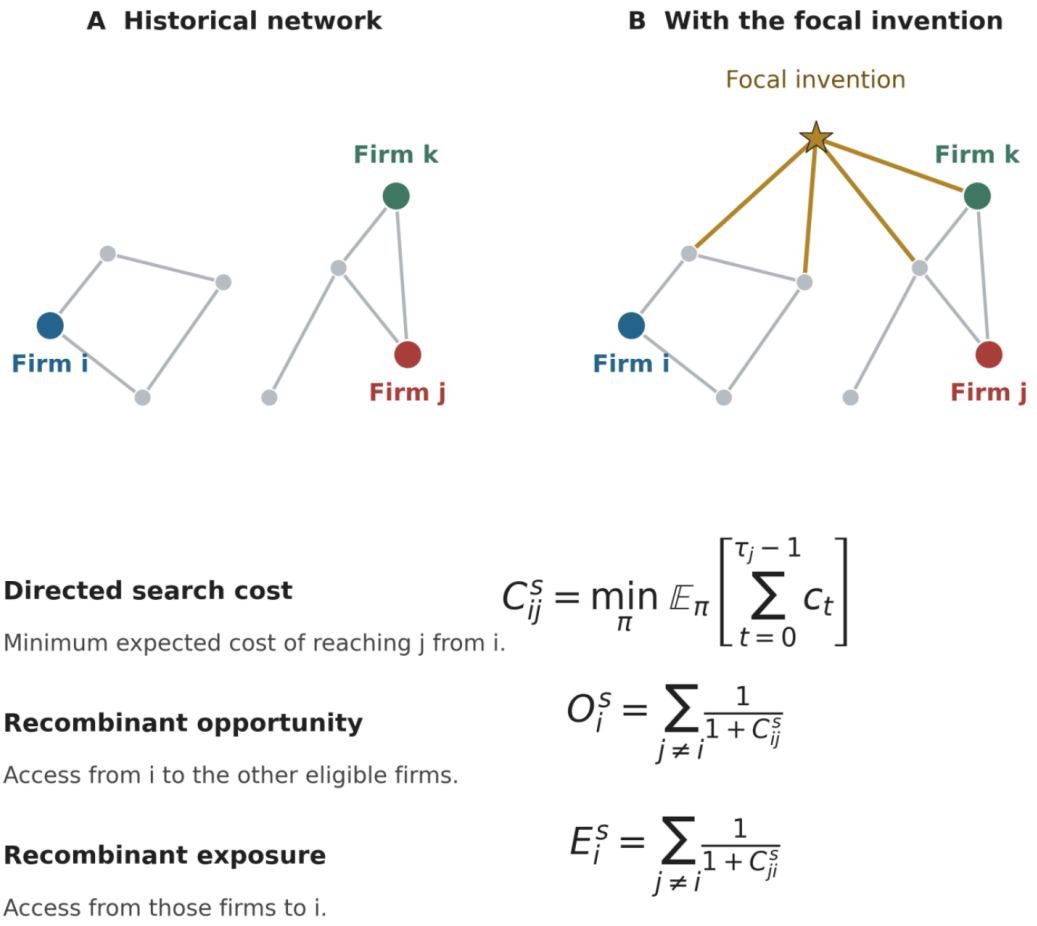


Figure 11: Patent-network intervention and directed search measures

Note. Nodes are patents and lines are citation connections, traversable in either direction. Colored nodes mark fixed firm representatives. The intervention adds only the focal invention and its eligible backward links. Search succeeds as intended with probability 0.90; errors follow the fixed historical-neighbor distribution. An optional random jump costs ten units. Opportunity and exposure sum opposite directions over the same firms. The network is schematic.

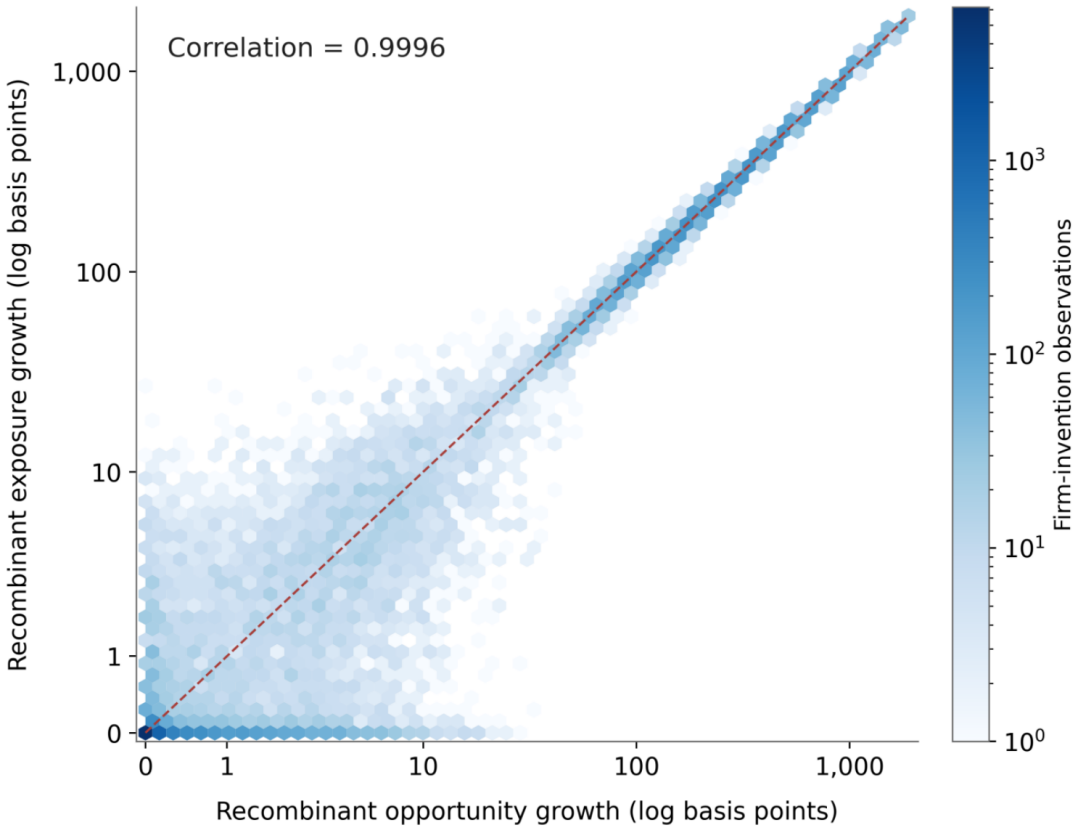


Figure 12: Changes in recombinant opportunity and exposure

Note. The plot contains 21,102 retained firm-invention observations. The dashed line marks equal growth in both directions. Growth is 10,000 times the log after/before ratio; axes use inverse-hyperbolic-sine spacing to display small and large changes together. Darker cells contain more observations.

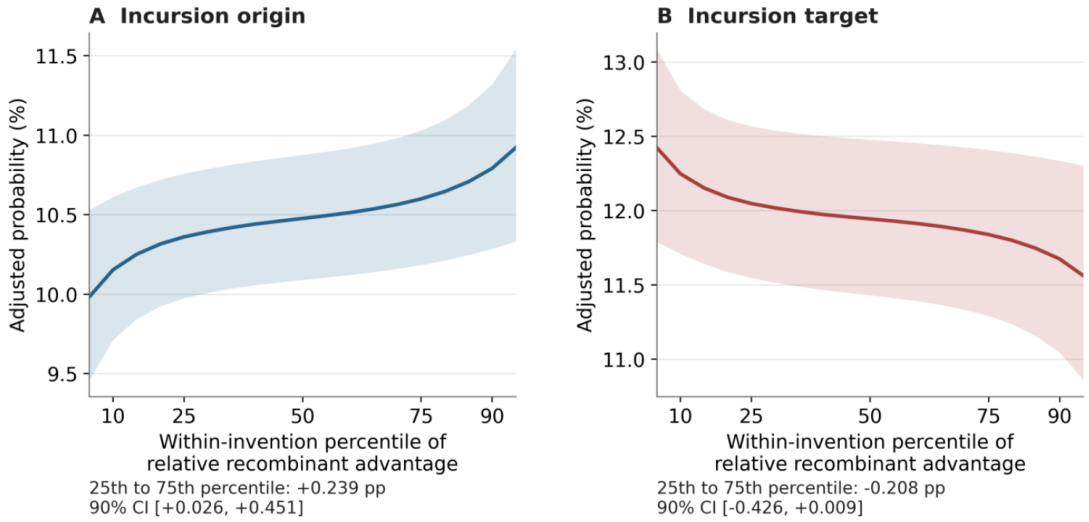


Figure 13: Relative recombinant advantage and firms' incursion roles

Note. Curves use the fully adjusted models in Table 9. At each percentile, firms receive the corresponding predictor value within their own focal invention; other covariates are held at observed values and predictions are averaged. Shading gives 90% pointwise confidence intervals, accounting for shared inventions and repeated firms. Probability contrasts are in percentage points (pp).

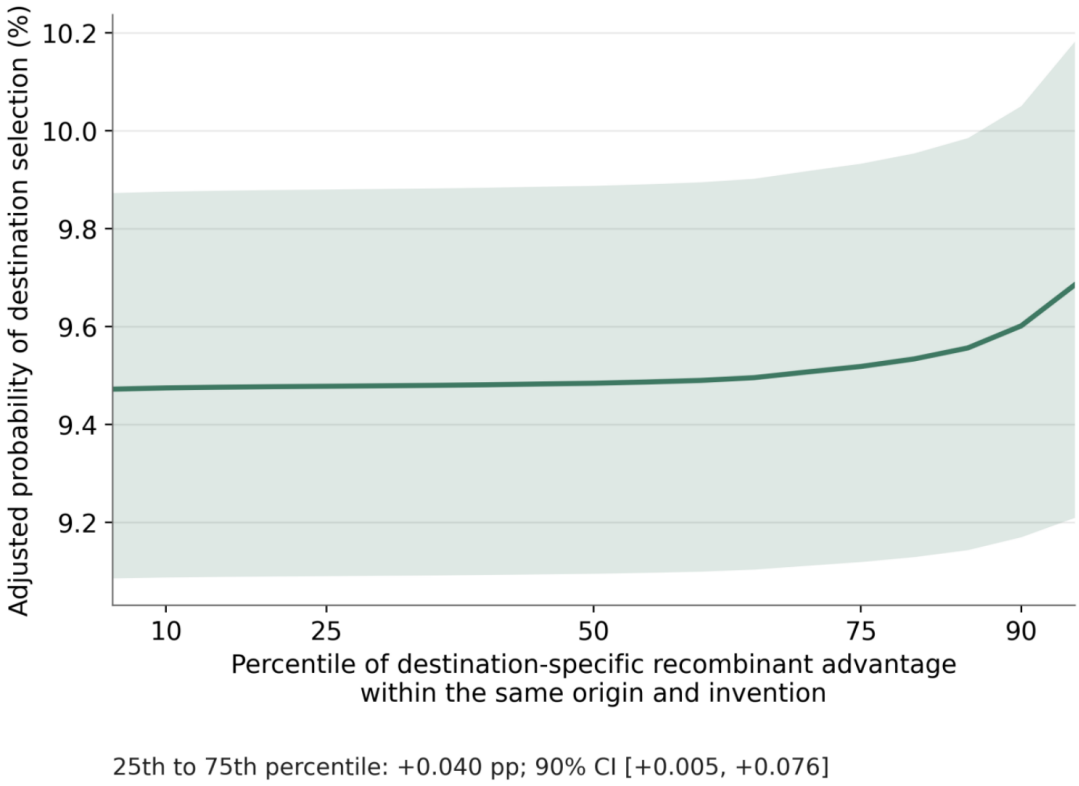


Figure 14: Destination-specific recombinant advantage and target selection

Note. Predictions use Table 9, Model 3, for 28,375 eligible pairs involving observed incursors. Predictor percentiles are calculated within each origin-invention group; covariates remain at observed values. Shading gives 90% pointwise intervals accounting for shared inventions and either firm. An origin can have more than one observed target.

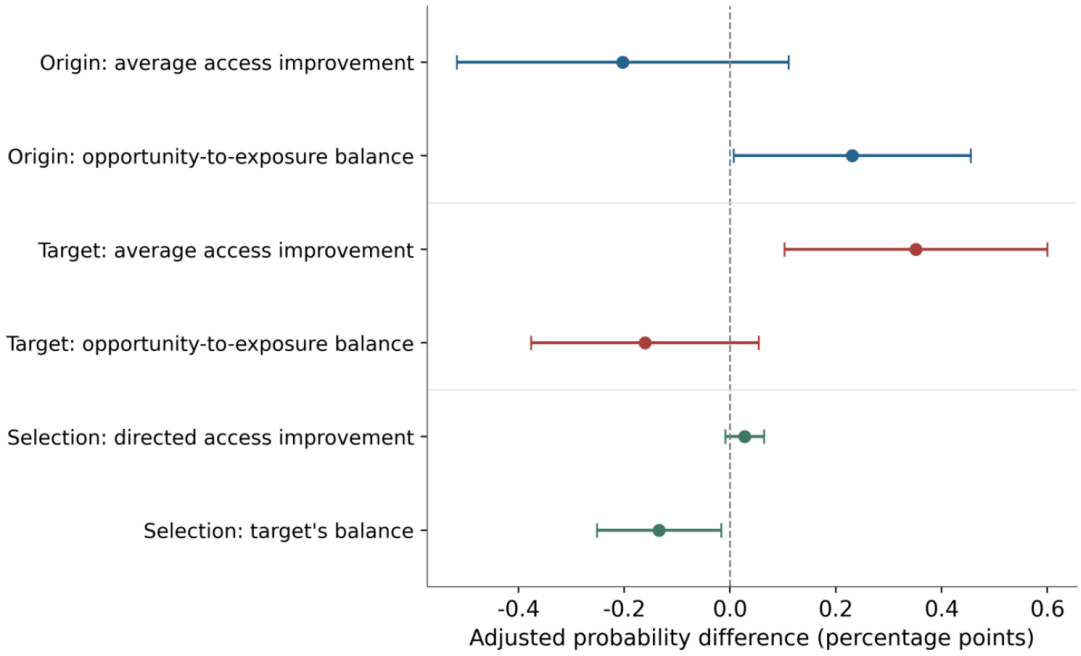


Figure 15: Probability contrasts from the component regressions

Note. Dots compare the within-group 25th and 75th percentiles of one component, holding the other component and controls at observed values. Bars are dependence-adjusted 90% confidence intervals. Firm models compare firms within inventions; destination selection compares targets of the same origin and invention. Components are entered jointly, as shown in Table 10.

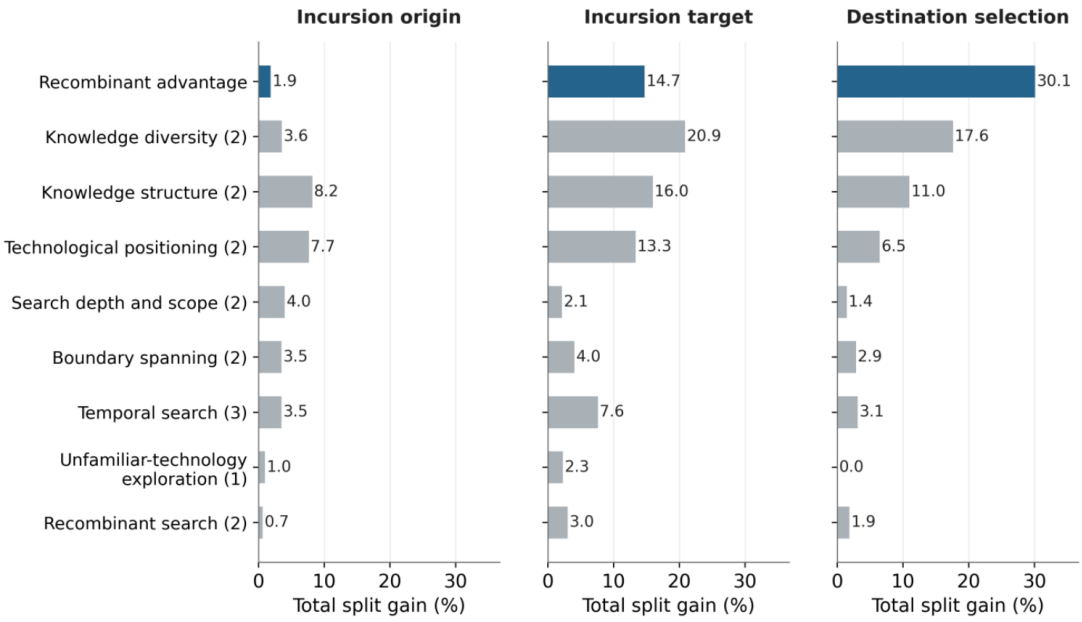


Figure 16: Grouped feature importance in the XGBoost models

Note. Bars show mean shares of total split gain across five fits separated by focal invention. The recombinant-advantage group contains three predictors in the firm models and eight in destination selection; counts for competing groups appear in parentheses. Basic controls are included in the denominator but not plotted. Split gain describes use in the fitted models, not effect direction, statistical significance, or out-of-sample accuracy gains.